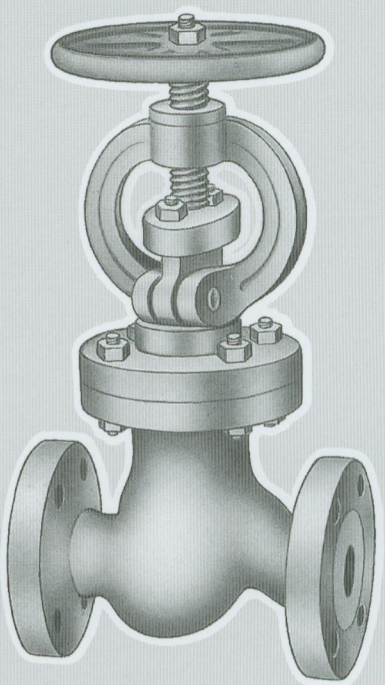


WORLD WIDE METRIC

# JIS

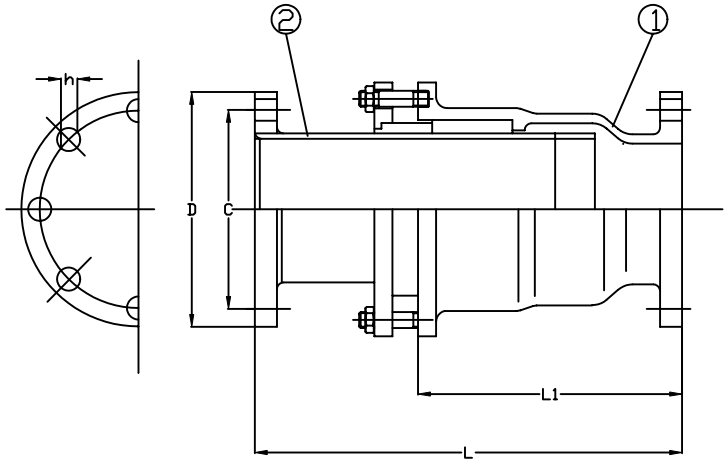
WORLD WIDE METRIC

# Table of Contents


| Page Nr. | JIS Nr. | Pressure | Description |
|---|---|---|---|
| 1 | Table Of Contents | | |
| | **JIS VALVE SECTION** | | |
| 2 | F3017 | 5K, 10K & 16K | Expansion Joint |
| 3 | F3059 | 5K | Storm Valve |
| 4 | F3059-10 | 10K | Storm Valve |
| 5 | F7301-F7303 | 5K & 16K | Globes |
| 6 | F7302-F7304 | 5K & 16K | Angles |
| 7 | F7305-F7353 | 5K | Globes |
| 8 | F7306-F7354 | 5K | Angles |
| 9 | F7307-F7375 | 10K | Globes |
| 10 | F7308-F7376 | 10K | Angles |
| 11 | F7309-F7377 | 16K | Globes |
| 12 | F7310-F7378 | 16K | Angles |
| 13 | F7311-F7312 | 5K | Globes |
| 14 | F7313-F7473 | 20K | Globes |
| 15 | F7314-F7474 | 20K | Angles |
| 16 | F7317-F7318 | 40K | Globes & Angles |
| 17 | F7319-F7471 | 10K | Globes |
| 18 | F7320-F7472 | 10K | Angles |
| 19 | F7329 | 40K | Globes |
| 20 | F7330 | 40K | Angles |
| 21 | F7334A | 5K & 10K | Hose Angles |
| 22 | F7334G | 5K & 10K | Hose Globes |
| 23 | F7343-F7389 | 20K | Cock & Screwed Angle |
| 24 | F7346-F7348 | 5K & 16K | Globes UB |
| 25 | F7347-F7349 | 5K & 16K | Angles UB |
| 26 | F7350 | 10K | Hull Angle Valve |
| 27 | F7351-F7409 | 5K & 16K | Globes SDNR |
| 28 | F7352-F7410 | 5K & 16K | Angles SDNR |
| 29 | F7356-F7371 | 5K | Lift & Swing Check |
| 30 | F7363 | 5K | Gate |
| 31 | F7363(S) | 5K | Gate |
| 32 | F7364 | 10K | Gate |
| 33 | F7366 | 10K | Gate |
| 34 | F7367-F7368 | 5K & 10K | Gate |
| 35 | F7369 | 16K | Gate |
| 36 | F7369-KOR | 16K | Gate |
| 37 | F7369S-KOR | 16K | Gate |
| 38 | F7372-F7373 | 5K & 10K | Swing Checks |
| 39 | F7381 | 5K | Cocks |
| 40 | F7387 | 16K | Cocks |
| 41 | F7388 | 20K | Globes |
| 42 | F7399 | 20K | Emergency Shut-off |
| 43 | F7415-F7418 | 5K & 16K | Lift Check |
| 44 | F7421 | 20K | Globes |
| 45 | F7422 | 20K | Angles |
| 46-47 | Butterfly Valves - Material Selection Guide | | |
| 48-49 | Butterfly Valves - Wafer, Lug and Flanged Patterns | | |
| | **JIS FLANGE SECTION** | | |
| 50 | B2220 | 5K and 10K | Slip-On & Blind |
| 51 | B2220 | 16K and 20K | Slip-On & Blind |

# SERIES F3017 - JIS 5K, 10K, and 16K
# Sleeve Type Expansion Joint

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 5K, 10K, or 16K




| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Ductile Iron | FCD45 |
| **2 - Sleeve** | Stainless Steel | SUS 304 |

| Nominal Pressure | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| | Body | Seat |
| 5K | 0.98 [10] | 0.78 [8] |
| 10K | 1.96 [20] | 1.57 [16] |
| 16K | 3.92 [40] | 3.14 [32] |

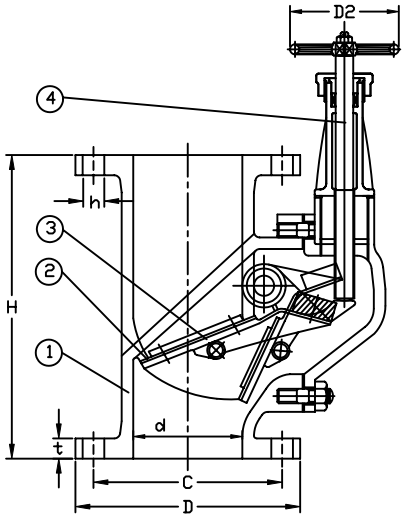
| | JIS 5K | | | | | | JIS 10K | | | | | | JIS 16K | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Size** | **L** | **L1** | **D** | **C** | **No.** | **h** | **L** | **L1** | **D** | **C** | **No.** | **h** | **L** | **L1** | **D** | **C** | **No.** | **h** |
| 50 | 345 | 208 | 130 | 105 | 4 | 15 | 368 | 229 | 155 | 120 | 4 | 19 | 432 | 251 | 155 | 120 | 8 | 19 |
| 65 | 350 | 213 | 155 | 130 | 4 | 15 | 382 | 236 | 175 | 140 | 4 | 19 | 456 | 259 | 175 | 140 | 8 | 19 |
| 80 | 385 | 243 | 180 | 145 | 4 | 19 | 415 | 277 | 185 | 150 | 8 | 19 | 469 | 295 | 200 | 160 | 8 | 23 |
| 100 | 390 | 249 | 200 | 165 | 8 | 19 | 415 | 277 | 210 | 175 | 8 | 19 | 476 | 295 | 225 | 185 | 8 | 23 |
| 125 | 390 | 249 | 235 | 200 | 8 | 19 | 432 | 282 | 250 | 210 | 8 | 23 | 520 | 300 | 270 | 225 | 8 | 25 |
| 150 | 400 | 254 | 265 | 230 | 8 | 19 | 439 | 287 | 280 | 240 | 8 | 23 | 522 | 305 | 305 | 260 | 12 | 25 |
| 200 | 420 | 282 | 320 | 280 | 8 | 23 | 458 | 306 | 330 | 290 | 12 | 23 | 544 | 327 | 350 | 305 | 12 | 25 |
| 250 | 435 | 292 | 385 | 345 | 12 | 23 | 475 | 311 | 400 | 355 | 12 | 25 | 576 | 332 | 430 | 380 | 12 | 27 |
| 300 | 455 | 305 | 430 | 390 | 12 | 23 | 510 | 338 | 445 | 400 | 16 | 25 | 588 | 362 | 480 | 430 | 16 | 27 |
| 350 | 465 | 310 | 480 | 435 | 12 | 25 | 517 | 343 | 490 | 445 | 16 | 25 | 612 | 367 | 540 | 480 | 16 | 33 |

# SERIES F3059 - JIS 5K
## Cast Steel Vertical Storm Valve

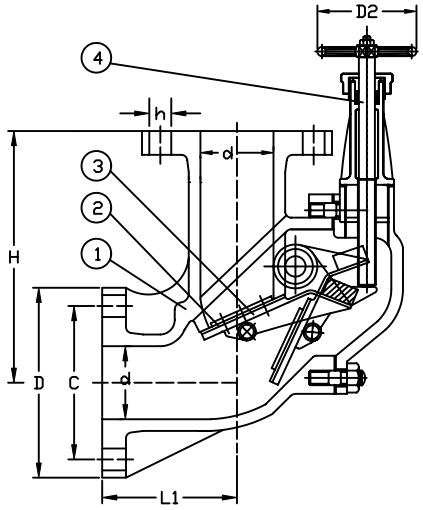
# SERIES F3060 - JIS 5K
## Cast Steel Angle Storm Valve

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 5K




Series F3059




Series F3060

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Seat** | Rubber | NBR |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Stainless Steel | SUS403 |

| Nominal Dia | Test pressure [kgf/cm2] |
|---|---|
| 50 to 200 | 5.1 |

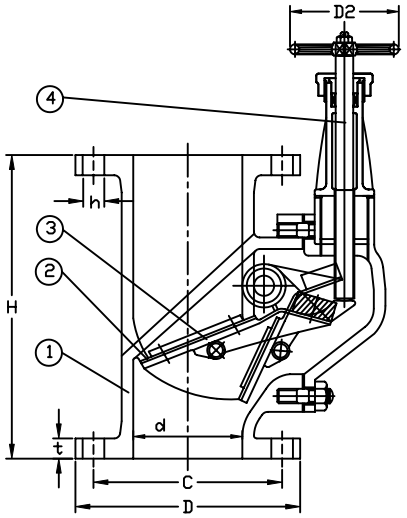
| Type | Size | H | D | C | No. | h | t | L1 | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|---|
| | 050 | 210 | 130 | 105 | 4 | 15 | 16 | - | 100 | 11.5 |
| | 065 | 240 | 155 | 130 | 4 | 15 | 18 | - | 100 | 16.5 |
| **F3059** | 080 | 260 | 180 | 145 | 4 | 19 | 18 | - | 125 | 21.6 |
| **Vertical** | 100 | 280 | 200 | 165 | 8 | 19 | 20 | - | 125 | 29.6 |
| | 125 | 330 | 235 | 200 | 8 | 19 | 20 | - | 125 | 43.6 |
| | 150 | 360 | 265 | 230 | 8 | 19 | 22 | - | 125 | 51.6 |
| | 200 | 459 | 320 | 280 | 8 | 23 | 24 | - | 125 | 106.5 |
| | 050 | 170 | 130 | 105 | 4 | 15 | 16 | 90 | 100 | 13 |
| | 065 | 200 | 155 | 130 | 4 | 15 | 18 | 110 | 100 | 19 |
| **F3060** | 080 | 220 | 180 | 145 | 4 | 19 | 18 | 120 | 125 | 23.8 |
| **Angle** | 100 | 250 | 200 | 165 | 8 | 19 | 20 | 130 | 125 | 34.5 |
| | 125 | 270 | 235 | 200 | 8 | 19 | 20 | 150 | 125 | 50 |
| | 150 | 310 | 265 | 230 | 8 | 19 | 22 | 170 | 125 | 67 |
| | 200 | 400 | 320 | 280 | 8 | 23 | 24 | 190 | 125 | 111 |

# SERIES F3059 - JIS 10K
## Cast Steel Vertical Storm Valve

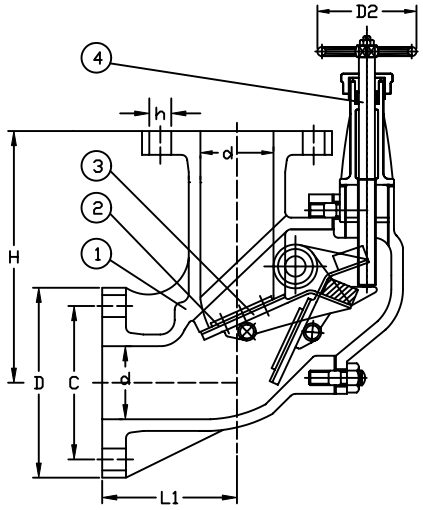
# SERIES F3060 - JIS 10K
## Cast Steel Angle Storm Valve

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220-10K




Series F3059




Series F3060

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Seat** | Rubber | NBR |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Stainless Steel | SUS403 |

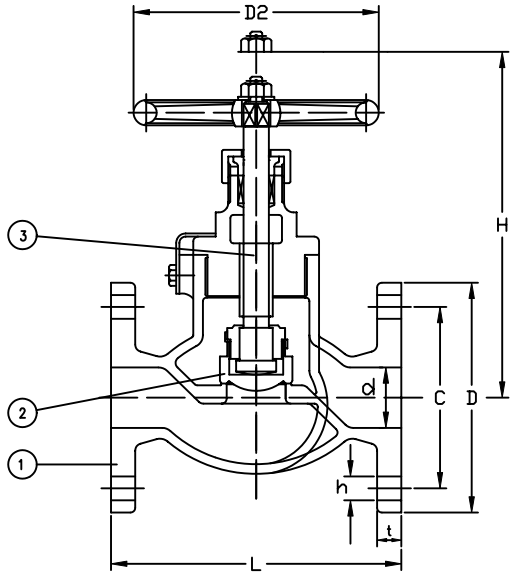
| Nominal Dia | Test pressure [kgf/cm2] |
|---|---|
| 50 to 200 | 5.1 |

| Type | Size | H | D | C | No. | h | t | L1 | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|---|
| | 050 | 210 | 155 | 120 | 4 | 19 | 16 | - | 100 | 11.5 |
| | 065 | 240 | 175 | 140 | 4 | 19 | 18 | - | 100 | 16.5 |
| **F3059** | 080 | 260 | 185 | 150 | 8 | 19 | 18 | - | 125 | 21.6 |
| **Vertical** | 100 | 280 | 210 | 175 | 8 | 19 | 18 | - | 125 | 29.6 |
| | 125 | 330 | 250 | 210 | 8 | 23 | 20 | - | 125 | 43.6 |
| | 150 | 360 | 280 | 240 | 8 | 23 | 22 | - | 125 | 51.6 |
| | 200 | 459 | 330 | 290 | 12 | 23 | 22 | - | 125 | 106.5 |
| | 050 | 170 | 155 | 120 | 4 | 19 | 16 | 90 | 100 | 13 |
| | 065 | 200 | 175 | 140 | 4 | 19 | 18 | 110 | 100 | 19 |
| **F3060** | 080 | 220 | 185 | 150 | 8 | 19 | 18 | 120 | 125 | 23.8 |
| **Angle** | 100 | 250 | 210 | 175 | 8 | 19 | 18 | 130 | 125 | 34.5 |
| | 125 | 270 | 250 | 210 | 8 | 23 | 20 | 150 | 125 | 50 |
| | 150 | 310 | 280 | 240 | 8 | 23 | 22 | 170 | 125 | 67 |
| | 200 | 400 | 330 | 290 | 12 | 23 | 22 | 190 | 125 | 111 |

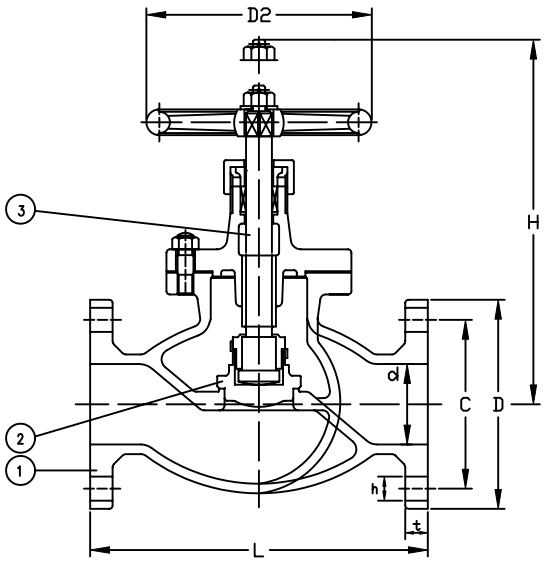
# SERIES F7301 - JIS 5K SERIES F7303 - JIS 16K

## Bronze Globe Valve

Manufactured in accordance to JIS F7400 Flanges as per JIS B2220 - 5K or 16K




Screwed Bonnet
for sizes 15A-40A




Bolted Bonnet
for sizes 50A-65A

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Stem** | Brass | C3771BD |

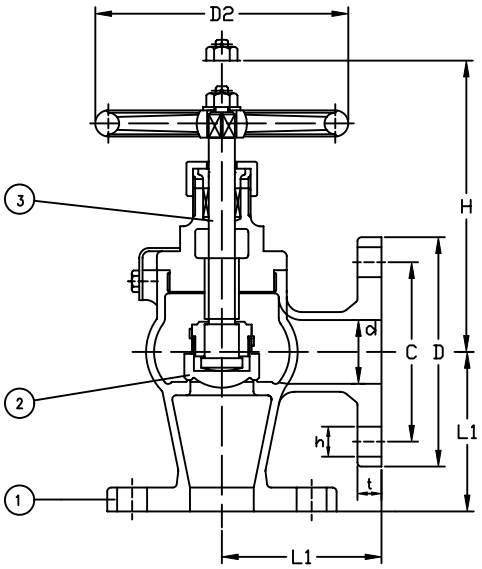
| Nominal Pressure | | 5K | 16K |
|---|---|---|---|
| Hydraulic | Body | 12kg | 38kg |
| Inspection | Seat | 8kg | 23kg |

| **Type** | **Size** | **L** | **D** | **C** | **No.** | **h** | **t** | **H** | **D2** |
|---|---|---|---|---|---|---|---|---|---|
| | 15 | 100 | 80 | 60 | 4 | 12 | 9 | 130 | 80 |
| | 20 | 110 | 85 | 65 | 4 | 12 | 10 | 140 | 100 |
| **F7301** | 25 | 120 | 95 | 75 | 4 | 12 | 10 | 160 | 125 |
| **5K** | 32 | 140 | 115 | 90 | 4 | 15 | 12 | 170 | 125 |
| | 40 | 160 | 120 | 95 | 4 | 15 | 12 | 190 | 140 |
| | 50 | 210 | 130 | 105 | 4 | 15 | 14 | 235 | 140 |
| | 65 | 250 | 155 | 130 | 4 | 15 | 14 | 260 | 160 |
| | 15 | 110 | 95 | 70 | 4 | 15 | 12 | 130 | 80 |
| **F7303** | 20 | 120 | 100 | 75 | 4 | 15 | 14 | 140 | 100 |
| **16K** | 25 | 130 | 125 | 90 | 4 | 19 | 14 | 160 | 125 |
| | 32 | 160 | 135 | 100 | 4 | 19 | 16 | 170 | 125 |
| | 40 | 180 | 140 | 105 | 4 | 19 | 16 | 190 | 140 |

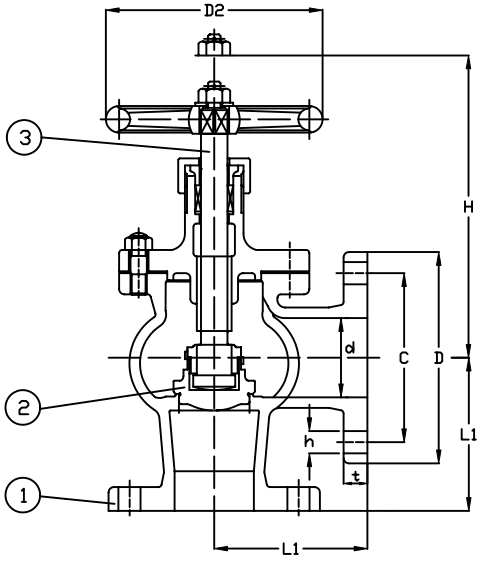
# SERIES F7302 - JIS 5K SERIES F7304 - JIS 16K

## Bronze Angle Valve

Manufactured in accordance to JIS F7400 Flanges as per JIS B2220 - 5K or 16K




Screwed Bonnet
for sizes 15A-40A




Bolted Bonnet
for sizes 50A-65A

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Stem** | Brass | C3771BD |

| Nominal Pressure | | 5K | 16K |
|---|---|---|---|
| Hydraulic | Body | 12kg | 38kg |
| Inspection | Seat | 8kg | 23kg |

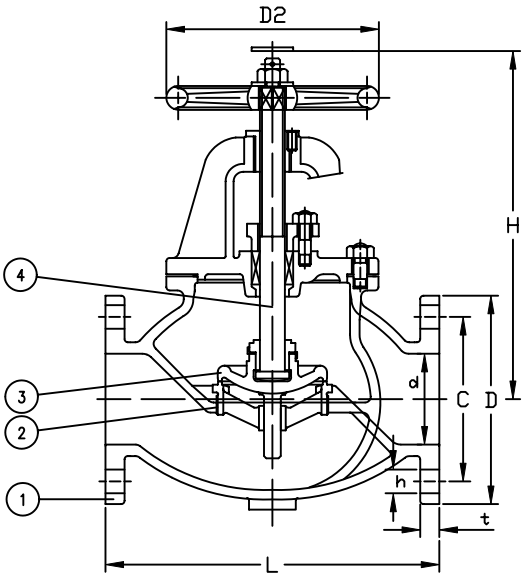
| Type | Size | L | D | C | No. | h | t | H | D2 |
|---|---|---|---|---|---|---|---|---|---|
| | 15 | 55 | 80 | 60 | 4 | 12 | 9 | 120 | 80 |
| | 20 | 60 | 85 | 65 | 4 | 12 | 10 | 130 | 100 |
| **F7302** | 25 | 65 | 95 | 75 | 4 | 12 | 10 | 145 | 125 |
| **5K** | 32 | 80 | 115 | 90 | 4 | 15 | 12 | 150 | 125 |
| | 40 | 85 | 120 | 95 | 4 | 15 | 12 | 165 | 140 |
| | 50 | 100 | 130 | 105 | 4 | 15 | 14 | 200 | 140 |
| | 65 | 115 | 155 | 130 | 4 | 15 | 14 | 220 | 160 |
| | 15 | 70 | 95 | 70 | 4 | 15 | 12 | 120 | 80 |
| **F7304** | 20 | 75 | 100 | 75 | 4 | 15 | 14 | 130 | 100 |
| **16K** | 25 | 85 | 125 | 90 | 4 | 19 | 14 | 145 | 125 |
| | 32 | 95 | 135 | 100 | 4 | 19 | 16 | 150 | 125 |
| | 40 | 100 | 140 | 105 | 4 | 19 | 16 | 165 | 140 |

# SERIES F7305 - JIS 5K
## Cast Iron Globe Valve

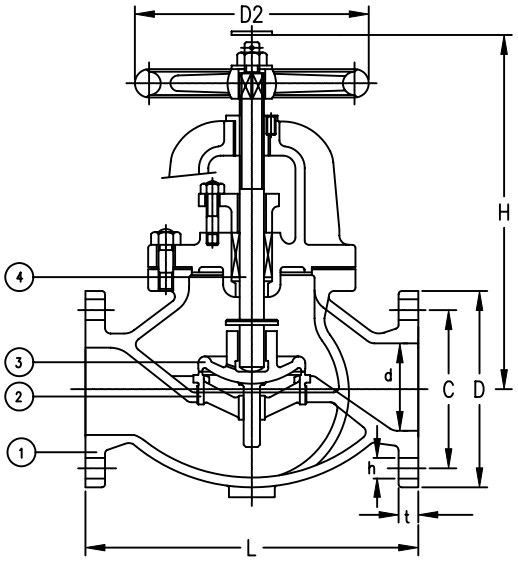
# SERIES F7353 - JIS 5K
## Cast Iron Globe Valve (SDNR)

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 -5K




Series F7305




Series F7353 -SDNR

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Seat** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

| Nominal | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| Diameter | Body | Seat |
| 300 or less | 1.03 [10.5] | 0.76 [7.7] |
| 350 and 400 | 0.88 [9] | 0.65 [6.6] |

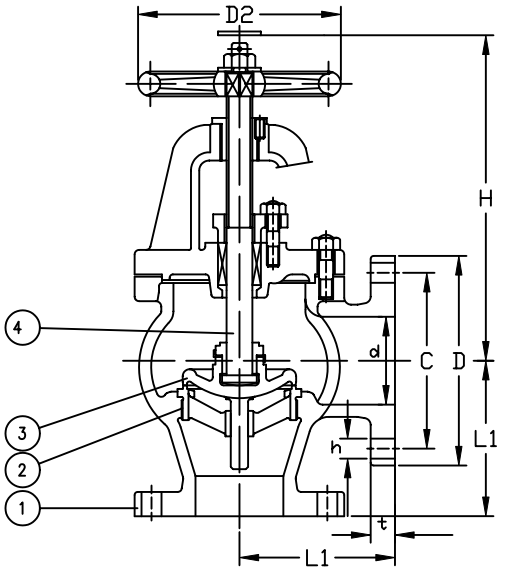
| **Size** | **L** | **D** | **C** | **No.** | **h** | **t** | **H** | **D2** | **Kg** |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 210 | 130 | 105 | 4 | 15 | 16 | 270 | 160 | 14.7 |
| 065 | 250 | 155 | 130 | 4 | 15 | 18 | 300 | 180 | 21.3 |
| 080 | 280 | 180 | 145 | 4 | 19 | 18 | 310 | 180 | 27.7 |
| 100 | 340 | 200 | 165 | 8 | 19 | 20 | 360 | 224 | 40.8 |
| 125 | 410 | 235 | 200 | 8 | 19 | 20 | 390 | 250 | 57.6 |
| 150 | 480 | 265 | 230 | 8 | 19 | 22 | 445 | 280 | 80.3 |
| 200 | 570 | 320 | 280 | 8 | 23 | 24 | 530 | 315 | 139 |
| 250 | 740 | 385 | 345 | 12 | 23 | 26 | 650 | 355 | 216 |
| 300 | 840 | 430 | 390 | 12 | 23 | 28 | 740 | 400 | 346 |
| 350 | 940 | 480 | 435 | 12 | 25 | 30 | 840 | 500 | 418 |
| 400 | 1050 | 540 | 495 | 16 | 25 | 30 | 940 | 630 | 600 |

# SERIES F7306 - JIS 5K
## Cast Iron Angle Valve

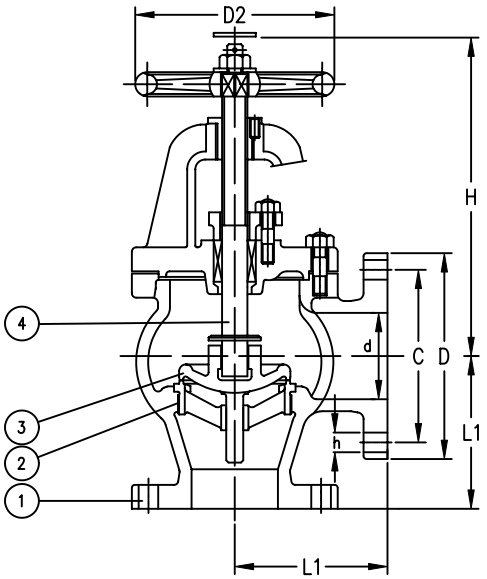
# SERIES F7354 - JIS 5K
## Cast Iron Angle Valve (SDNR)

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 5K




Series F7306




Series F7354 -SDNR

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Seat** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

| Nominal Diameter | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| | Body | Seat |
| 300 or less | 1.03 [10.5] | 0.76 [7.7] |
| 350 and 400 | 0.88 [9] | 0.65 [6.6] |
| 450 and over | 0.74 [7.5] | 0.54 [5.5] |

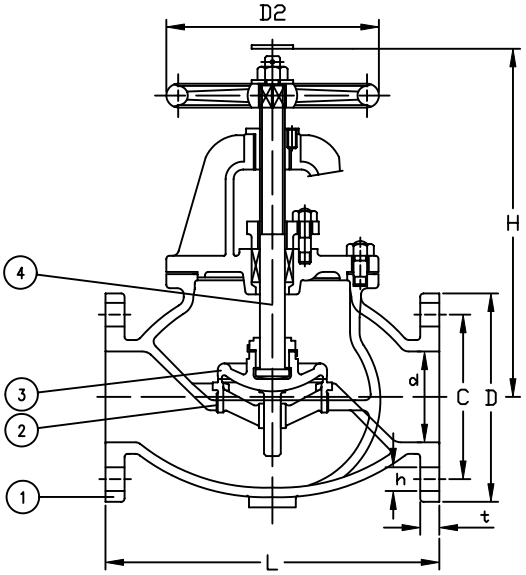
| **Size** | **L1** | **D** | **C** | **No.** | **h** | **t** | **H** | **D2** | **Kg** |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 100 | 130 | 105 | 4 | 15 | 16 | 240 | 160 | 12.9 |
| 065 | 115 | 155 | 130 | 4 | 15 | 18 | 255 | 180 | 18.2 |
| 080 | 130 | 180 | 145 | 4 | 19 | 18 | 265 | 180 | 23.4 |
| 100 | 150 | 200 | 165 | 8 | 19 | 20 | 310 | 224 | 33.7 |
| 125 | 170 | 235 | 200 | 8 | 19 | 20 | 330 | 250 | 46.2 |
| 150 | 190 | 265 | 230 | 8 | 19 | 22 | 380 | 280 | 63.4 |
| 200 | 220 | 320 | 280 | 8 | 23 | 24 | 450 | 315 | 105 |
| 250 | 275 | 385 | 345 | 12 | 23 | 26 | 540 | 355 | 166 |
| 300 | 310 | 430 | 390 | 12 | 23 | 28 | 610 | 400 | 243 |
| 350 | 360 | 480 | 435 | 12 | 25 | 30 | 690 | 500 | 327 |
| 400 | 395 | 540 | 495 | 16 | 25 | 30 | 770 | 630 | 459 |
| 450 | 440 | 605 | 555 | 16 | 25 | 30 | 835 | 630 | 618 |
| 500 | 485 | 655 | 605 | 20 | 25 | 32 | 915 | 710 | 800 |

# SERIES F7307 - JIS 10K
## Cast Iron Globe Valve

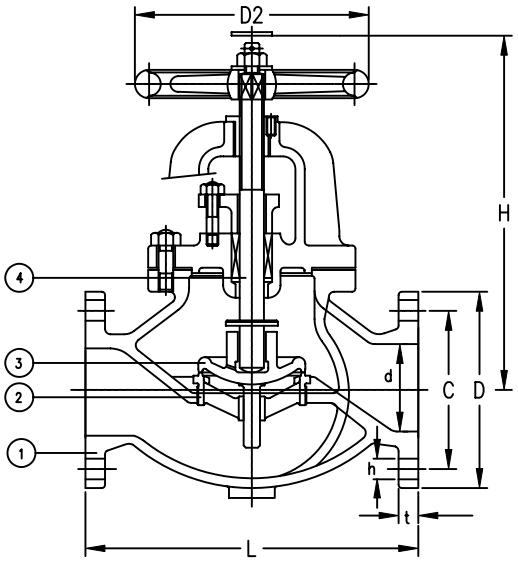
# SERIES F7375 - JIS 10K
## Cast Iron Globe Valve (SDNR)

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 10K




Series F7307




Series F7375 - SDNR

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Seat** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

| Nominal Diameter | Test Pressure MPa (kgf/cm2) | |
|---|---|---|
| | Body | Seat |
| 200 or less | 2.06 [21] | 1.51 [15.4] |
| 250 or more | 1.47 [15] | 1.08 [11] |

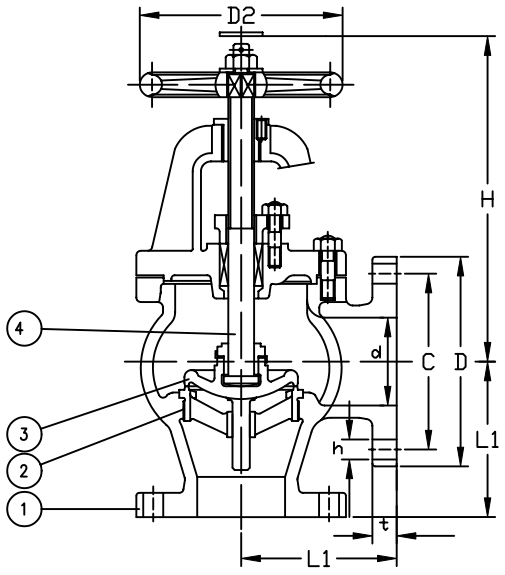
| **Size** | **L** | **D** | **C** | **No.** | **h** | **t** | **H** | **D2** | **Kg** |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 220 | 155 | 120 | 4 | 19 | 20 | 285 | 160 | 19.9 |
| 065 | 270 | 175 | 140 | 4 | 19 | 22 | 310 | 200 | 28.7 |
| 080 | 300 | 185 | 150 | 8 | 19 | 22 | 320 | 200 | 32.4 |
| 100 | 350 | 210 | 175 | 8 | 19 | 24 | 370 | 250 | 48.7 |
| 125 | 420 | 250 | 210 | 8 | 23 | 24 | 420 | 280 | 73.1 |
| 150 | 490 | 280 | 240 | 8 | 23 | 26 | 470 | 315 | 104 |
| 200 | 570 | 330 | 290 | 12 | 23 | 26 | 555 | 355 | 162 |
| 250 | 740 | 400 | 355 | 12 | 25 | 30 | 680 | 450 | 304 |
| 300 | 840 | 445 | 400 | 16 | 25 | 32 | 770 | 500 | 458 |

# SERIES F7308 - JIS 10K
## Cast Iron Angle Valve

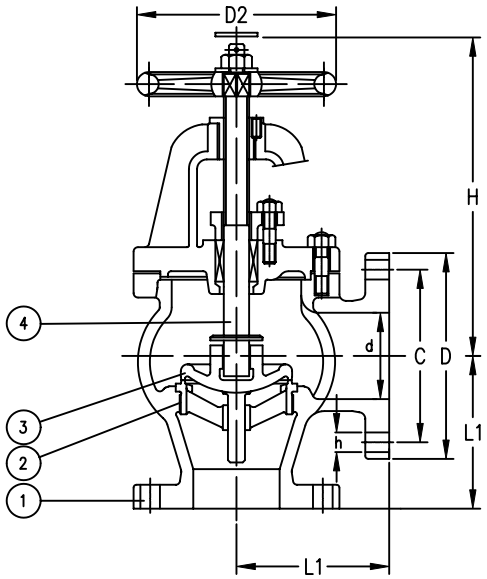
# SERIES F7376 - JIS 10K
## Cast Iron Angle Valve (SDNR)

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 -10K




Series F7308




Series F7376- SDNR

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Seat** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

| Nominal Diameter | Test Pressure MPa (kgf/cm2) | |
|---|---|---|
| | Body | Seat |
| 200 or less | 2.06 [21] | 1.51 [15.4] |
| 250 or more | 1.47 [15] | 1.08 [11] |

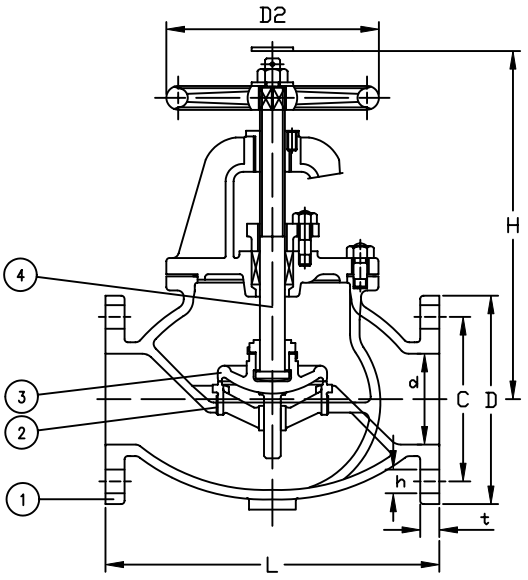
| Size | L1 | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 120 | 155 | 120 | 4 | 19 | 20 | 255/255 | 160 | 18.2 |
| 065 | 130 | 175 | 140 | 4 | 19 | 22 | 270/260 | 200 | 25.4 |
| 080 | 140 | 185 | 150 | 8 | 19 | 22 | 275/270 | 200 | 27.4 |
| 100 | 160 | 210 | 175 | 8 | 19 | 24 | 315/305 | 250 | 40.6 |
| 125 | 180 | 250 | 210 | 8 | 23 | 24 | 360/350 | 280 | 69.9 |
| 150 | 205 | 280 | 240 | 8 | 23 | 26 | 405/390 | 315 | 81 |
| 200 | 230 | 330 | 290 | 12 | 23 | 26 | 475/450 | 355 | 125 |
| 250 | 290 | 400 | 355 | 12 | 25 | 30 | 570 | 450 | 211 |
| 300 | 320 | 445 | 400 | 16 | 25 | 32 | 645 | 500 | 293 |
| 350 | 360 | 490 | 445 | 16 | 25 | 34 | 710 | 560 | 386 |
| 400 | 420 | 560 | 510 | 16 | 27 | 36 | 790 | 630 | 544 |

# SERIES F7309 - JIS 16K
## Cast Iron Globe Valve

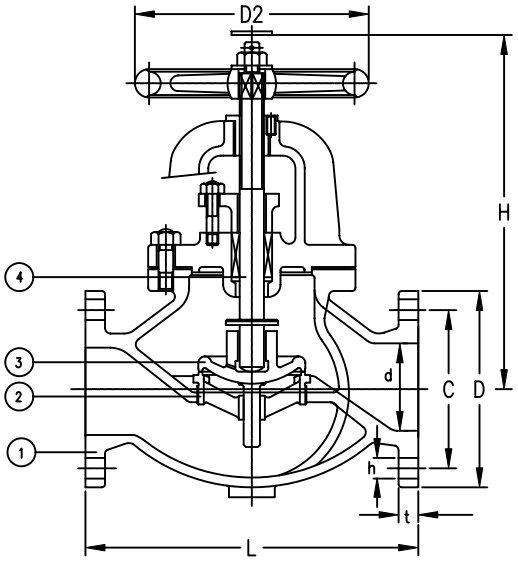
# SERIES F7377 - JIS 16K
## Cast Iron Globe Valve (SDNR)

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 16K




Series F7309




Series F7377 - SDNR

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Seat** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

| Nominal Diameter | Test Pressure MPa (kgf/cm2) | |
|---|---|---|
| | Body | Seat |
| 50 to 200 | 3.2 [33] | 2.38 [24.2] |

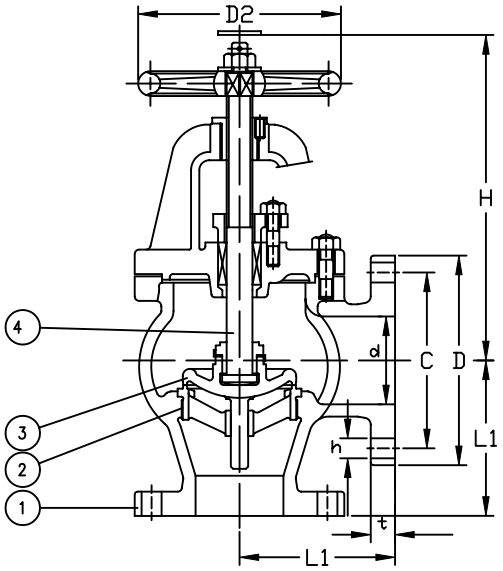
| **Size** | **L** | **D** | **C** | **No.** | **h** | **t** | **H** | **D2** | **Kg** |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 220 | 155 | 155 | 8 | 19 | 20 | 285 | 160 | 19.6 |
| 065 | 270 | 175 | 175 | 8 | 19 | 22 | 310 | 200 | 28.3 |
| 080 | 300 | 200 | 200 | 8 | 23 | 24 | 340 | 224 | 40.4 |
| 100 | 350 | 225 | 225 | 8 | 23 | 26 | 385 | 250 | 56.9 |
| 125 | 430 | 270 | 270 | 8 | 25 | 26 | 455 | 315 | 89.5 |
| 150 | 500 | 305 | 305 | 12 | 25 | 28 | 510 | 355 | 123 |
| 200 | 570 | 350 | 350 | 12 | 25 | 30 | 630 | 450 | 188 |

# SERIES F7310 - JIS 16K
## Cast Iron Angle Valve

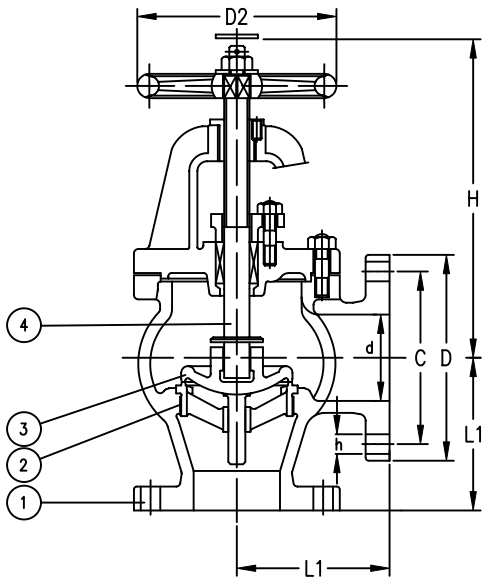
# SERIES F7378 - JIS 16K
## Cast Iron Angle Valve (SDNR)

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 16K




Series F7310




Series F7378 - SDNR

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Seat** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

| Nominal Diameter | Test Pressure MPa (kgf/cm2) | |
|---|---|---|
| | Body | Seat |
| 50 to 200 | 3.2 [33] | 2.38 [24.2] |

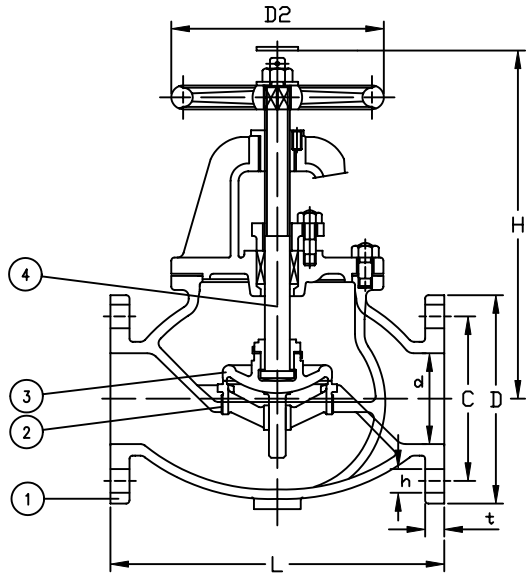
| **Size** | **L1** | **D** | **C** | **No.** | **h** | **t** | **H** | **D2** | **Kg** |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 120 | 155 | 120 | 8 | 19 | 20 | 255 | 160 | 18 |
| 065 | 130 | 175 | 140 | 8 | 19 | 22 | 270 | 200 | 25 |
| 080 | 150 | 200 | 160 | 8 | 23 | 24 | 295 | 224 | 35.7 |
| 100 | 170 | 225 | 185 | 8 | 23 | 26 | 330 | 250 | 49.2 |
| 125 | 200 | 270 | 225 | 8 | 25 | 26 | 390 | 315 | 76.1 |
| 150 | 225 | 305 | 260 | 12 | 25 | 28 | 435 | 355 | 103 |
| 200 | 250 | 350 | 305 | 12 | 25 | 30 | 540 | 450 | 155 |

# SERIES F7311 - JIS 5K
## Cast Steel Globe Valve

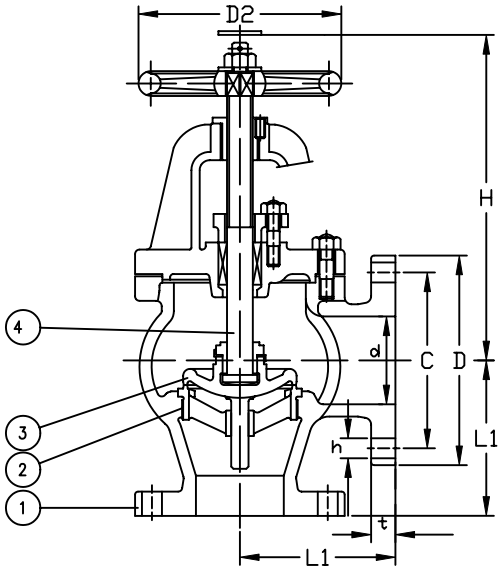
# SERIES F7312 - JIS 5K
## Cast Steel Angle Valve

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 5K




Series F7311




Series F7312

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Steel | SC46 |
| **2 - Seat** | Stainless Steel | SCS2 |
| **3 - Disc** | Stainless Steel | SCS2 |
| **4 - Stem** | Stainless Steel | SUS403 |

| | Test Pressure MPa [kg/cm2] |
|---|---|
| Body | 1.04 [10.5] |
| Seat | 0.76 [7.7] |

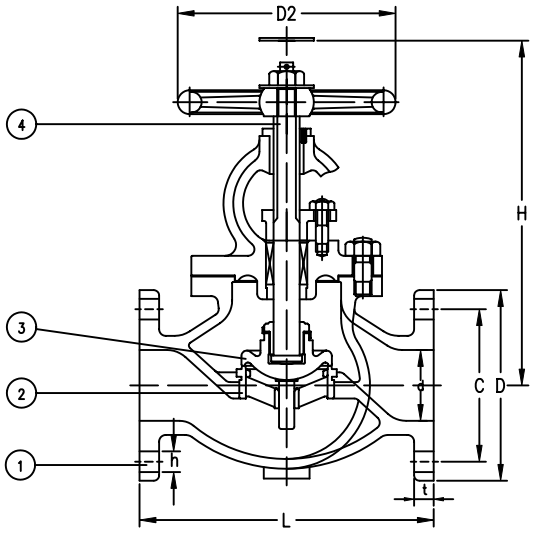
| | JIS F7311 | | | JIS F7312 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Size** | **L** | **H** | **Kg** | **L1** | **H** | **Kg** | **D** | **C** | **No.** | **h** | **t** | **D2** |
| 050 | 210 | 270 | 15.5 | 100 | 240 | 13.3 | 130 | 105 | 4 | 15 | 14 | 160 |
| 065 | 250 | 300 | 25.8 | 115 | 255 | 19.5 | 155 | 130 | 4 | 15 | 14 | 180 |
| 080 | 280 | 310 | 32.3 | 130 | 265 | 25.5 | 180 | 145 | 4 | 19 | 14 | 180 |
| 100 | 340 | 360 | 47.6 | 150 | 310 | 34.8 | 200 | 165 | 8 | 19 | 16 | 224 |
| 125 | 410 | 390 | 71.3 | 170 | 330 | 46.5 | 235 | 200 | 8 | 19 | 16 | 250 |
| 150 | 480 | 445 | 86 | 190 | 380 | 66.5 | 265 | 230 | 8 | 19 | 18 | 280 |
| 200 | 570 | 530 | 147 | 220 | 450 | 111 | 320 | 280 | 8 | 23 | 20 | 315 |
| 250 | 740 | 791 | 230 | 275 | 641 | 178 | 385 | 345 | 12 | 23 | 22 | 355 |
| 300 | 840 | 929 | 309 | 310 | 748 | 275 | 430 | 390 | 12 | 23 | 22 | 400 |

# SERIES F7313 - JIS 20K
## Cast Steel Globe Valve

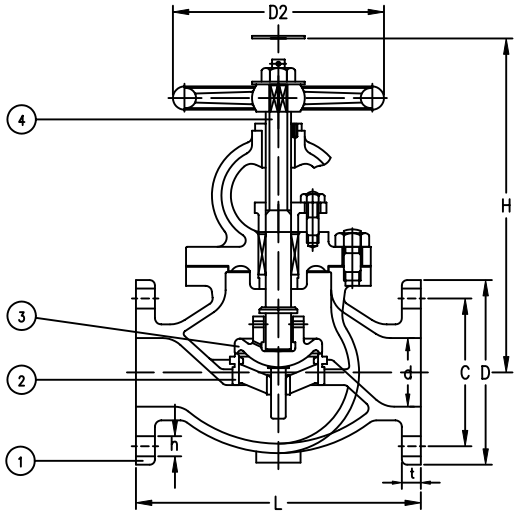
# SERIES F7473 - JIS 20K
## Cast Steel Globe Valve (SDNR)

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 20K




Series F7313




Series F7473 - SDNR

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Seat** | Stainless Steel | SCS2 |
| **3 - Disc** | Stainless Steel | SCS2 |
| **4 - Stem** | Stainless Steel | SUS403 |

| | Test pressure MPa [kgf/cm2] |
|---|---|
| Body | 5.00 [51.0] |
| Seat | 3.67 [37.4] |

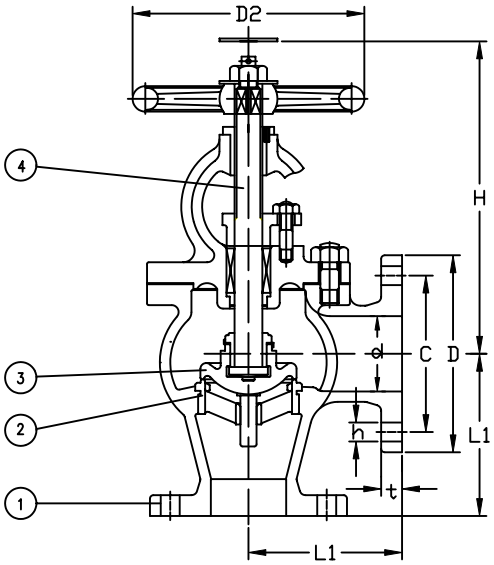
| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 032 | 190 | 135 | 100 | 4 | 19 | 18 | 270 | 160 | 14.4 |
| 040 | 200 | 140 | 105 | 4 | 19 | 18 | 190 | 160 | 16 |
| 050 | 230 | 155 | 120 | 8 | 19 | 18 | 305 | 200 | 21.6 |
| 065 | 270 | 175 | 140 | 8 | 19 | 20 | 345 | 224 | 30.4 |
| 080 | 300 | 200 | 160 | 8 | 23 | 22 | 385 | 250 | 43.5 |
| 100 | 350 | 225 | 185 | 8 | 23 | 24 | 440 | 280 | 62.3 |
| 125 | 430 | 270 | 225 | 8 | 25 | 26 | 500 | 315 | 94.8 |
| 150 | 500 | 305 | 260 | 12 | 25 | 28 | 550 | 355 | 132 |
| 200 | 560 | 350 | 305 | 12 | 25 | 30 | 630 | 450 | 204 |
| 250 | 660 | 430 | 380 | 12 | 27 | 34 | 725 | 560 | 336 |

# SERIES F7314 - JIS 20K
## Cast Steel Angle Valve

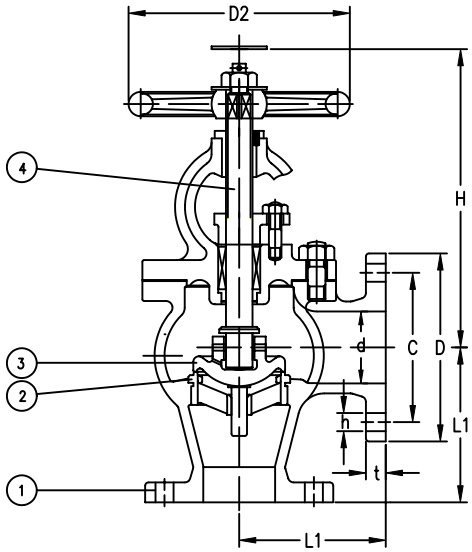
# SERIES F7474 - JIS 20K
## Cast Steel Angle Valve (SDNR)

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 20K




Series F7314




Series F7474 - SDNR

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Seat** | Stainless Steel | SCS2 |
| **3 - Disc** | Stainless Steel | SCS2 |
| **4 - Stem** | Stainless Steel | SUS403 |

| | Test pressure MPa [kgf/cm2] |
|---|---|
| Body | 5.00 [51.0] |
| Seat | 3.67 [37.4] |

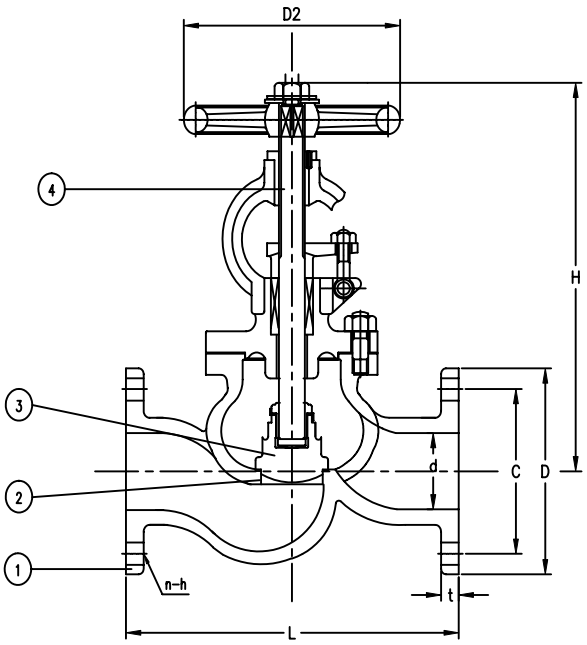
| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 032 | 100 | 135 | 100 | 4 | 19 | 18 | 250 | 160 | 13 |
| 040 | 110 | 140 | 105 | 4 | 19 | 18 | 265 | 160 | 15.2 |
| 050 | 125 | 155 | 120 | 8 | 19 | 18 | 275 | 200 | 20 |
| 065 | 135 | 175 | 140 | 8 | 19 | 20 | 305 | 224 | 26.9 |
| 080 | 150 | 200 | 160 | 8 | 23 | 22 | 340 | 250 | 38.9 |
| 100 | 170 | 225 | 185 | 8 | 23 | 24 | 385 | 280 | 54.4 |
| 125 | 200 | 270 | 225 | 8 | 25 | 26 | 440 | 315 | 79.9 |
| 150 | 225 | 305 | 260 | 12 | 25 | 28 | 485 | 355 | 111 |
| 200 | 280 | 350 | 305 | 12 | 25 | 30 | 540 | 450 | 181 |
| 250 | 310 | 430 | 380 | 12 | 27 | 34 | 620 | 560 | 283 |

# SERIES F7317 - JIS 40K
## Cast Steel Globe Valve

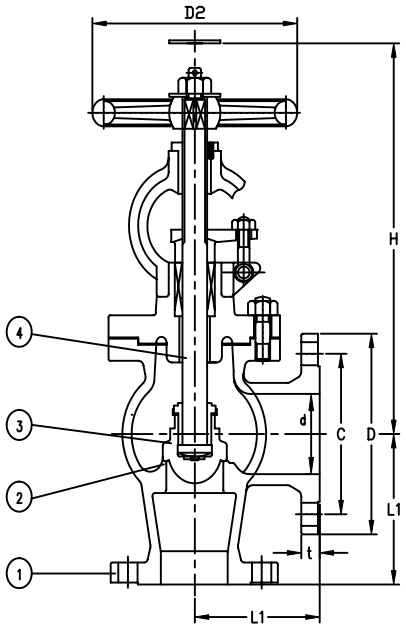
# SERIES F7318 - JIS 40K
## Cast Steel Angle Valve

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 40K




Series F7317




Series F7318

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Stem** | Stainless Steel | SUS403 |
| **3 - Disc** | Stainless Steel | SCS2 |
| **4 - Seat** | Stainless Steel | SCS2 |

| | Test pressure MPa [kgf/cm2] |
|---|---|
| Body | 9.12 [93.0] |
| Seat | 6.69 [68.2] |

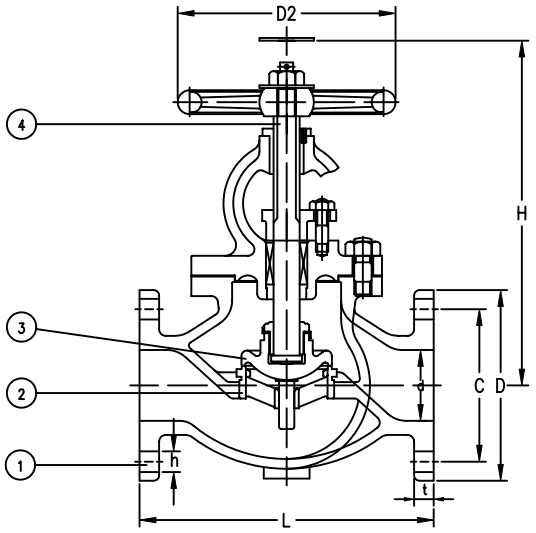
| | JIS F7317 | | | JIS F7318 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Size** | **L** | **H** | **KG** | **L1** | **H** | **Kg** | **D** | **C** | **No.** | **h** | **t** | **D2** |
| 032 | 229 | 370 | 25.4 | 114 | 350 | 24.9 | 140 | 105 | 4 | 19 | 24 | 200 |
| 040 | 241 | 395 | 30.2 | 121 | 370 | 29.4 | 160 | 120 | 4 | 23 | 24 | 200 |
| 050 | 292 | 435 | 41.0 | 146 | 405 | 40.0 | 165 | 130 | 8 | 19 | 26 | 224 |
| 065 | 330 | 490 | 62.3 | 165 | 455 | 60.3 | 200 | 160 | 8 | 23 | 30 | 250 |
| 080 | 356 | 540 | 79.0 | 178 | 500 | 76.2 | 210 | 170 | 8 | 23 | 32 | 315 |
| 100 | 432 | 640 | 140 | 216 | 590 | 135 | 250 | 205 | 8 | 25 | 36 | 400 |
| 125 | 508 | 720 | 197 | 254 | 660 | 189 | 300 | 250 | 8 | 27 | 40 | 450 |
| 150 | 559 | 815 | 298 | 279 | 740 | 284 | 355 | 295 | 12 | 33 | 44 | 500 |

# SERIES F7319 - JIS 10K
## Cast Steel Globe Valve

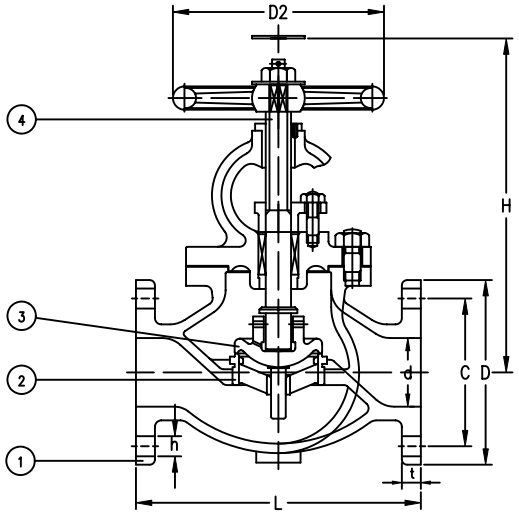
# SERIES F7471 - JIS 10K
## Cast Steel Globe Valve (SDNR)

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 10K




Series F7319




Series F7471 - SDNR

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Seat** | Stainless Steel | SCS2 |
| **3 - Disc** | Stainless Steel | SCS2 |
| **4 - Stem** | Stainless Steel | SUS403 |

| | Test pressure MPa [kgf/cm2] |
|---|---|
| Body | 2.06 [21.0] |
| Seat | 1.51 [15.4] |

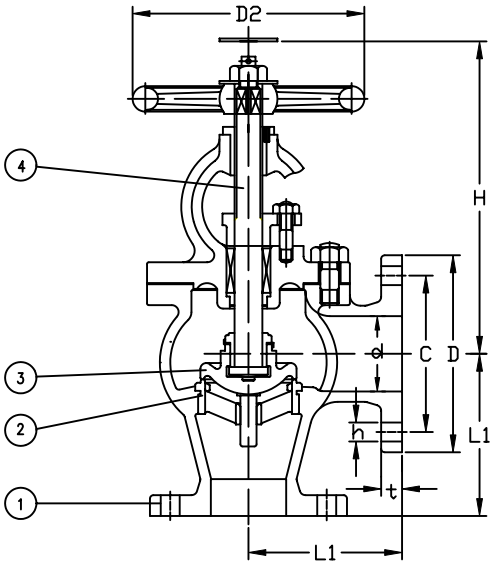
| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 220 | 155 | 120 | 4 | 19 | 20 | 285 | 160 | 19.9 |
| 065 | 270 | 175 | 140 | 4 | 19 | 22 | 310 | 200 | 28.7 |
| 080 | 300 | 185 | 150 | 8 | 19 | 22 | 320 | 200 | 32.4 |
| 100 | 350 | 210 | 175 | 8 | 19 | 24 | 370 | 250 | 48.7 |
| 125 | 420 | 250 | 210 | 8 | 23 | 24 | 420 | 280 | 73.1 |
| 150 | 490 | 280 | 240 | 8 | 23 | 26 | 470 | 315 | 104 |
| 200 | 570 | 330 | 290 | 12 | 23 | 26 | 555 | 355 | 162 |
| 250 | 740 | 400 | 355 | 12 | 25 | 30 | 680 | 450 | 304 |
| 300 | 840 | 445 | 400 | 16 | 25 | 32 | 770 | 500 | 458 |

# SERIES F7320 - JIS 10K
## Cast Steel Angle Valve

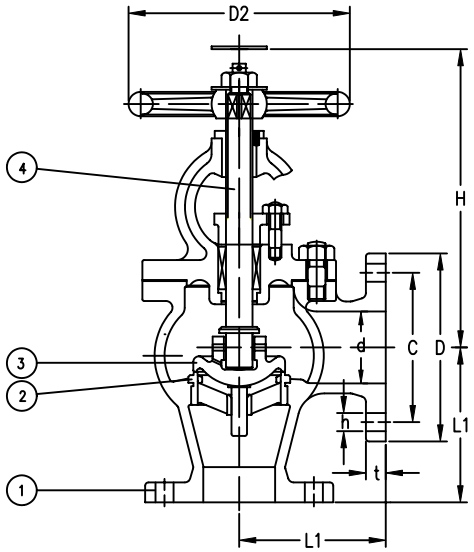
# SERIES F7472 - JIS 10K
## Cast Steel Angle Valve (SDNR)

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 10K




Series F7320




Series F7472

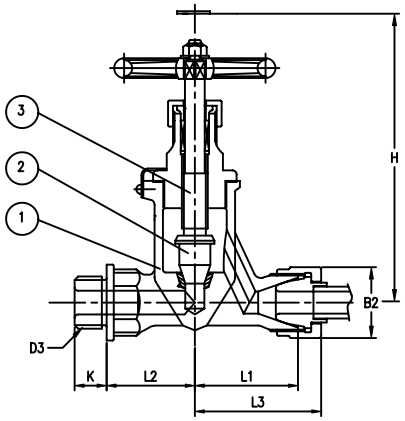
| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Seat** | Stainless Steel | SCS2 |
| **3 - Disc** | Stainless Steel | SCS2 |
| **4 - Stem** | Stainless Steel | SUS403 |

| | Test pressure MPa [kgf/cm2] |
|---|---|
| Body | 2.06 [21.0] |
| Seat | 1.51 [15.4] |

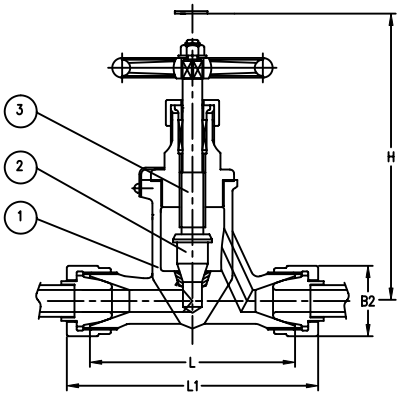
| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 120 | 155 | 120 | 4 | 19 | 16 | 240 | 160 | 16.3 |
| 065 | 130 | 175 | 140 | 4 | 19 | 18 | 260 | 200 | 23 |
| 080 | 140 | 185 | 150 | 8 | 19 | 18 | 265 | 200 | 25.3 |
| 100 | 160 | 210 | 175 | 8 | 19 | 18 | 300 | 250 | 37 |
| 125 | 180 | 250 | 210 | 8 | 23 | 20 | 350 | 280 | 57.1 |
| 150 | 205 | 280 | 240 | 8 | 23 | 22 | 400 | 315 | 77.2 |
| 200 | 230 | 330 | 290 | 12 | 23 | 22 | 480 | 355 | 116 |
| 250 | 290 | 400 | 355 | 12 | 25 | 24 | 535 | 400 | 188 |
| 300 | 320 | 445 | 400 | 16 | 25 | 24 | 610 | 450 | 256 |

# SERIES F7329 - JIS 40K
# Forged Steel Globe Valve

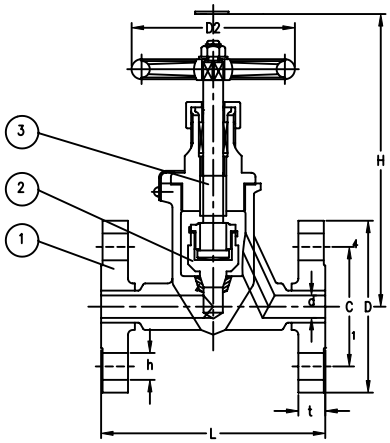
Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - JIS 40K




"S" Type




"U" Type




"F" Type

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Forged Steel | SF45 |
| **2 - Stem** | Stainless Steel | SUS403 |
| **3 - Disc** | Stainless Steel | SUS402J1 |

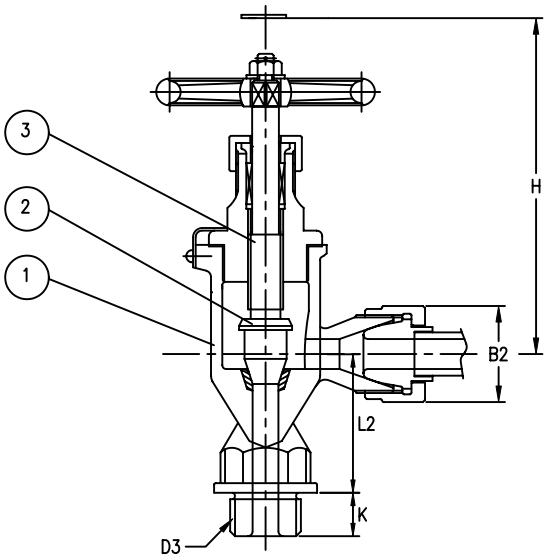
| | Test Pressure MPa [kg/cm2] |
|---|---|
| Body | 9.12 [93.0] |
| Seat | 6.69 [68.2] |

| Type | Type of Connection | Nominal Dia. |
|---|---|---|
| S | Screw thread type | 6 and 10 |
| U | Union type | 6 and 10 |
| F | Flange type | 15 to 25 |

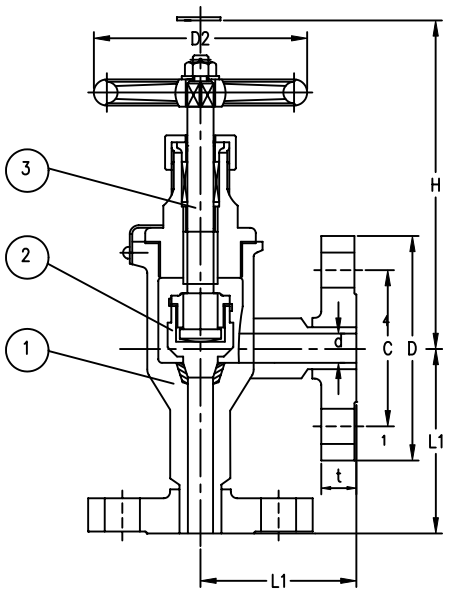
| | Size | L | L1 | L2 | L3 | K | D | C | n | h | t | H | B2 | D3 | Kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Type | 6 | - | 42 | 38 | 51 | 12 | - | - | - | - | - | 150 | 30 | M20x1.5 | 1.5 |
| "S" | 10 | - | 51 | 45 | 63 | 16 | - | - | - | - | - | 160 | 41 | M42x2 | 1.8 |
| Type | 6 | 84 | 102 | - | - | - | - | - | - | - | - | 150 | 30 | - | 1.54 |
| "U" | 10 | 102 | 126 | - | - | - | - | - | - | - | - | 160 | 41 | - | 1.91 |
| Type | 15 | 150 | - | - | - | - | 115 | 80 | 4 | 19 | 20 | 215 | - | - | 5.91 |
| "F" | 20 | 170 | - | - | - | - | 120 | 85 | 4 | 19 | 20 | 250 | - | - | 7.4 |
| | 25 | 200 | - | - | - | - | 130 | 95 | 4 | 19 | 22 | 270 | - | - | 9.79 |

# SERIES F7330 - JIS 40K
# Forged Steel Angle Valve

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 40K




"S" Type




"F" Type

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Forged Steel | SF45 |
| **2 - Stem** | Stainless Steel | SUS403 |
| **3 - Disc** | Stainless Steel | SUS402J1 |

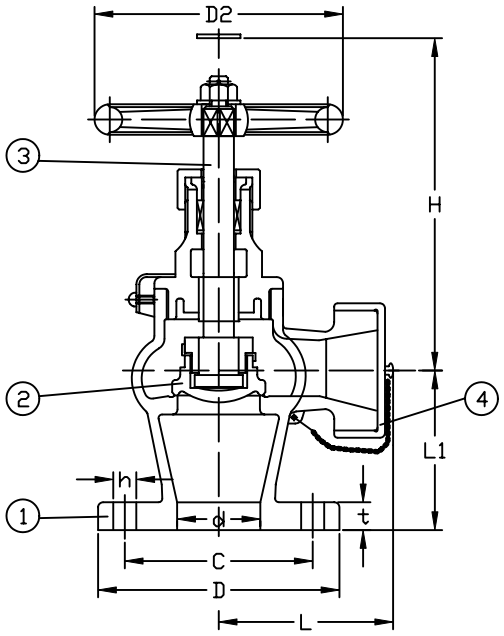
| | Test Pressure MPa [kg/cm2] |
|---|---|
| Body | 9.12 [93.0] |
| Seat | 6.69 [68.2] |

| Type | Type of Connection | Nominal Dia. |
|---|---|---|
| S | Screw thread type | 6 and 10 |
| F | Flange type | 15 to 25 |

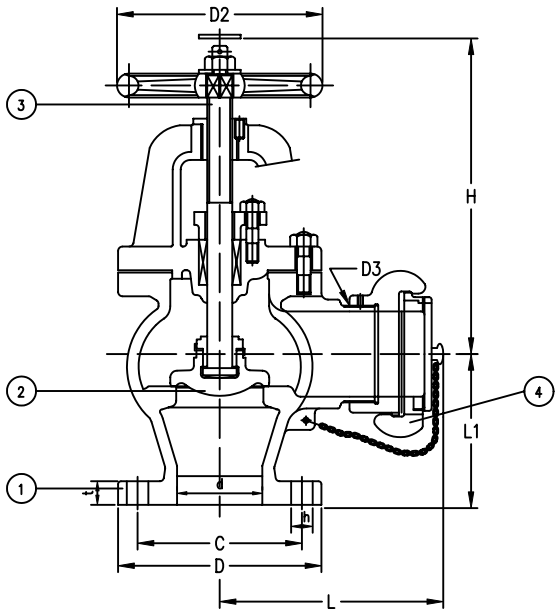
| | Size | L | L1 | L2 | L3 | K | D | C | n | h | t | H | B2 | D3 | Kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Type | 6 | - | 42 | 38 | 51 | 12 | - | - | - | - | - | 140 | 30 | M20x1.5 | 1.47 |
| "S" | 10 | - | 46 | 45 | 58 | 16 | - | - | - | - | - | 145 | 41 | M24x2 | 1.7 |
| Type | 15 | 90 | - | - | - | - | 115 | 80 | 4 | 19 | 20 | 190 | - | - | 5.89 |
| "F" | 20 | 95 | - | - | - | - | 120 | 85 | 4 | 19 | 20 | 220 | - | - | 7.27 |
| | 25 | 100 | - | - | - | - | 130 | 95 | 4 | 19 | 22 | 235 | - | - | 9.38 |

# SERIES F7334 - JIS 5K & 10K Bronze Hose Angle Valve
# Nakajima System

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 5K or 10K




Screwed Bonnet
for sizes 15A-25A




Bolted Bonnet
for sizes 40A-65A

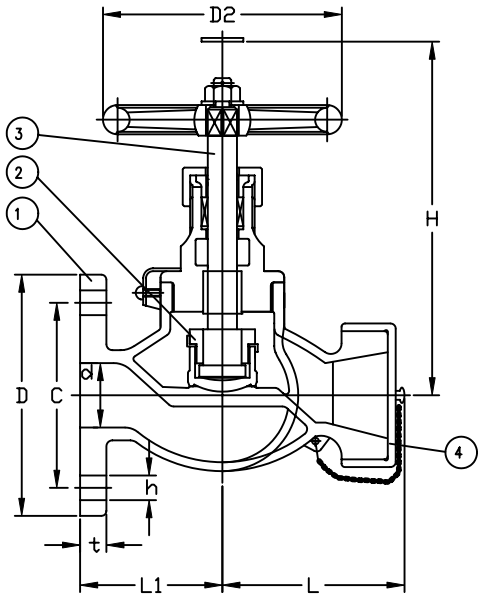
| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Stem** | Brass | C3771BD |
| **4 - Coupling** | Bronze | BC6 |

| Nominal Pressure | | 5K | 10K |
|---|---|---|---|
| Hydraulic | Body | 12kg | 24kg |
| Inspection | Seat | 8kg | 12kg |

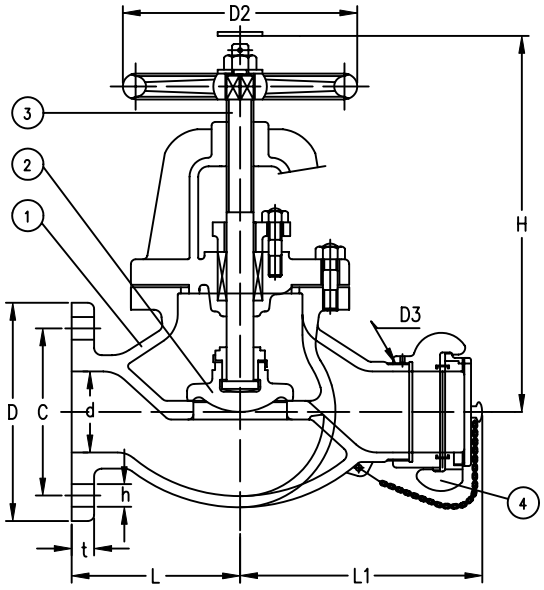
| Flange | Size | L | L1 | D | C | No. | h | t | H | D2 | D3 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **5K** | 15 | 68 | 55 | 80 | 60 | 4 | 12 | 9 | 129 | 80 | M42x2 |
| | 25 | 78 | 65 | 95 | 75 | 4 | 12 | 10 | 161 | 125 | M42x2 |
| | 40 | 124 | 85 | 120 | 95 | 4 | 15 | 12 | 193 | 140 | M52x2 |
| | 50 | 155 | 100 | 130 | 105 | 4 | 15 | 14 | 256 | 160 | M64x2 |
| **10K** | 15 | 73 | 70 | 95 | 70 | 4 | 15 | 12 | 129 | 80 | M42x2 |
| | 25 | 83 | 85 | 125 | 90 | 4 | 19 | 14 | 161 | 125 | M42x2 |
| | 40 | 124 | 100 | 140 | 105 | 4 | 19 | 16 | 193 | 140 | M52x2 |
| | 50 | 160 | 120 | 155 | 120 | 4 | 19 | 16 | 271 | 160 | M64x2 |
| | 65 | 180 | 130 | 175 | 140 | 4 | 19 | 18 | 298 | 200 | M80x2 |

# SERIES F7334 - JIS 5K & 10K Bronze Hose Globe Valve
# Nakajima System

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 5K or 10K




Screwed Bonnet
for sizes 15A-25A




Bolted Bonnet
for sizes 40A-65A

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Stem** | Brass | C3771BD |
| **4 - Coupling** | Bronze | BC6 |

| Nominal Pressure | | 5K | 10K |
|---|---|---|---|
| Hydraulic | Body | 12kg | 24kg |
| Inspection | Seat | 8kg | 12kg |

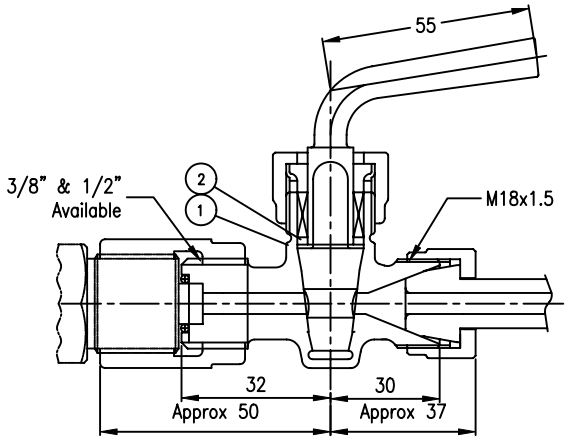
| Type | Size | L | L1 | D | C | No. | h | t | H | D2 | D3 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **5K** | 15 | 73 | 50 | 80 | 60 | 4 | 12 | 9 | 129 | 80 | M42x2 |
| | 25 | 78 | 60 | 95 | 75 | 4 | 12 | 10 | 161 | 125 | M42x2 |
| | 40 | 128 | 80 | 120 | 95 | 4 | 15 | 12 | 193 | 140 | M52x2 |
| | 50 | 160 | 105 | 130 | 105 | 4 | 15 | 14 | 256 | 160 | M64x2 |
| **10K** | 15 | 73 | 55 | 95 | 70 | 4 | 15 | 12 | 129 | 80 | M42x2 |
| | 25 | 83 | 65 | 125 | 90 | 4 | 19 | 14 | 161 | 125 | M42x2 |
| | 40 | 133 | 90 | 140 | 105 | 4 | 19 | 16 | 193 | 140 | M52x2 |
| | 50 | 165 | 110 | 155 | 120 | 4 | 19 | 16 | 271 | 160 | M64x2 |
| | 65 | 200 | 135 | 175 | 140 | 4 | 19 | 18 | 298 | 200 | M80x2 |

# SERIES F7343 - JIS 20K
# Bronze Guage Cock

Manufactured in accordance to JIS F7400

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Plug** | Bronze | BC6 |

| | | |
|---|---|---|
| Hydraulic | Body | 30kg |
| Inspection | Seat | 22kg |

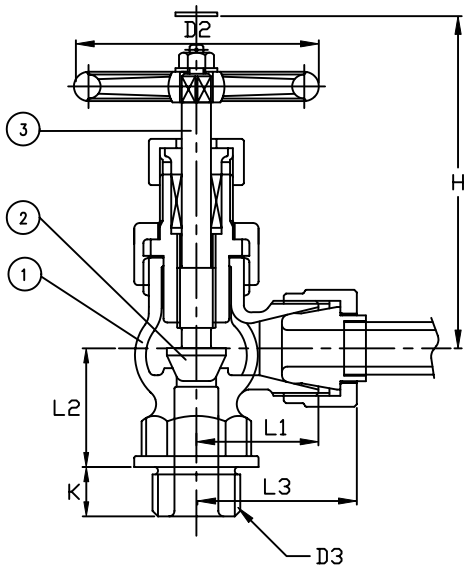



# SERIES F7389 - JIS 20K
# Bronze Angle Valve

Manufactured in accordance to JIS F7400

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Stem** | Bronze | C3771BD |

| | | |
|---|---|---|
| Hydraulic | Body | 48kg |
| Inspection | Seat | 29kg |

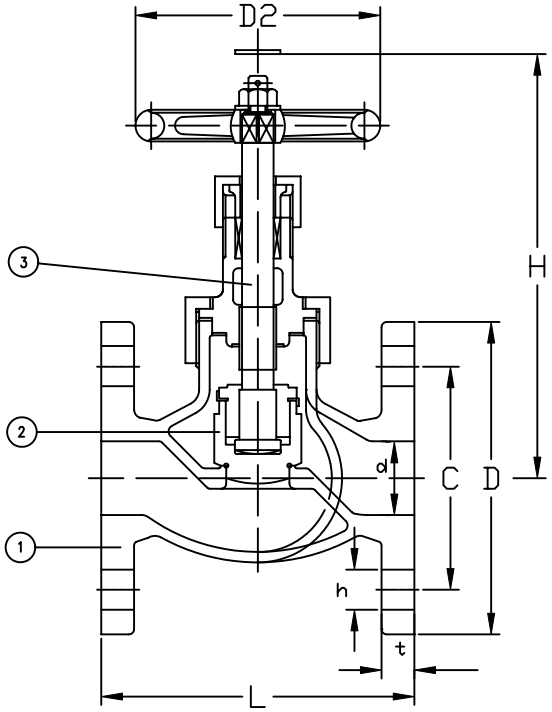



| Size | Pipe Out Dia Copper | Steel | L1 | L2 | L3 | d3 | K | H | D2 |
|---|---|---|---|---|---|---|---|---|---|
| 6 | 8 / 10 | 10.5 | 32 | 32 | 41 | M20x1.5 | 12 | 95 | 63 |
| 10 | 15 | 17.3 | 39 | 35 | 51 | M24x2 | 16 | 105 | 80 |

# SERIES F7346 - JIS 5K SERIES F7348 - JIS 16K

## Bronze Globe Valve -Union Bonnet

Manufactured in accordance to JIS F7400 Flanges as per JIS B2220 - 5K or 16K




| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Stem** | Brass | C3771BD |

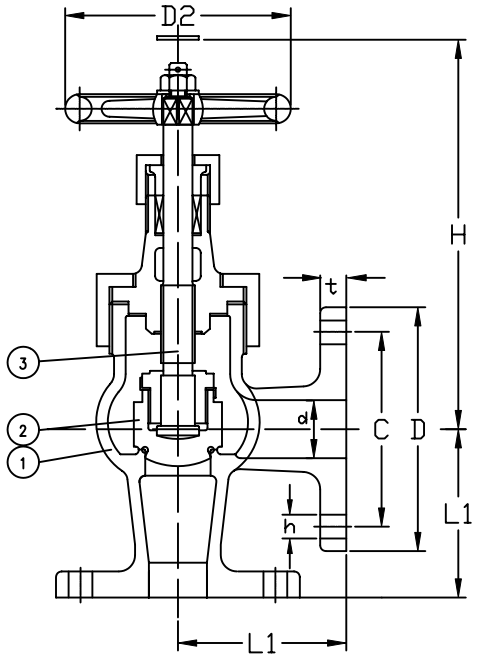
| Nominal Pressure | | 5K | 16K |
|---|---|---|---|
| Hydraulic | Body | 12kg | 38kg |
| Inspection | Seat | 8kg | 23kg |

| Type | Size | L | D | C | No. | h | t | H | D2 |
|---|---|---|---|---|---|---|---|---|---|
| | 15 | 100 | 80 | 60 | 4 | 12 | 9 | 155 | 80 |
| | 20 | 110 | 85 | 65 | 4 | 12 | 10 | 165 | 100 |
| **F7346** | 25 | 120 | 95 | 75 | 4 | 12 | 10 | 185 | 125 |
| **5K** | 32 | 140 | 115 | 90 | 4 | 15 | 12 | 195 | 125 |
| | 40 | 160 | 120 | 95 | 4 | 15 | 12 | 210 | 140 |
| | 15 | 110 | 95 | 70 | 4 | 15 | 12 | 155 | 80 |
| **F7348** | 20 | 120 | 100 | 75 | 4 | 15 | 14 | 165 | 100 |
| **16K** | 25 | 130 | 125 | 90 | 4 | 19 | 14 | 185 | 125 |
| | 32 | 160 | 135 | 100 | 4 | 19 | 16 | 195 | 125 |
| | 40 | 180 | 140 | 105 | 4 | 19 | 16 | 210 | 140 |

# SERIES F7347 - JIS 5K SERIES F7349 - JIS 16K

## Bronze Angle Valve - Union Bonnet

Manufactured in accordance to JIS F7400 Flanges as per JIS B2220 - 5K or 16K



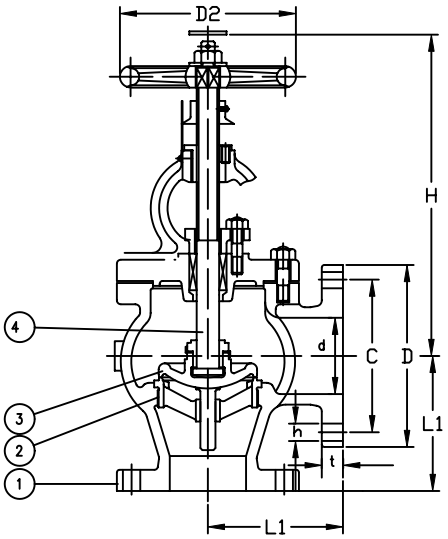

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Stem** | Brass | C3771BD |

| Nominal Pressure | | 5K | 16K |
|---|---|---|---|
| Hydraulic | Body | 12kg | 38kg |
| Inspection | Seat | 8kg | 23kg |

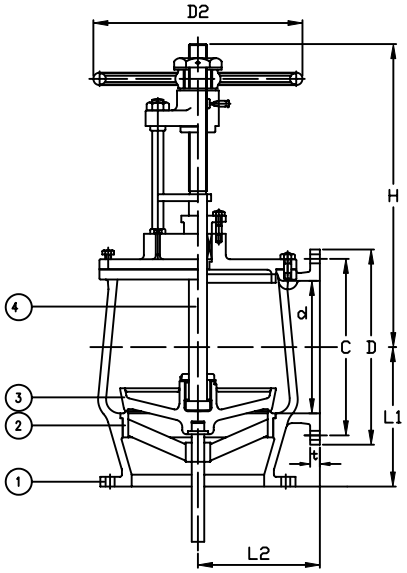
| Type | Size | L | D | C | No. | h | t | H | D2 |
|---|---|---|---|---|---|---|---|---|---|
| | 15 | 55 | 80 | 60 | 4 | 12 | 9 | 145 | 80 |
| | 20 | 60 | 85 | 65 | 4 | 12 | 10 | 155 | 100 |
| **F7347** | 25 | 65 | 95 | 75 | 4 | 12 | 10 | 175 | 125 |
| **5K** | 32 | 80 | 115 | 90 | 4 | 15 | 12 | 175 | 125 |
| | 40 | 85 | 120 | 95 | 4 | 15 | 12 | 190 | 140 |
| | 15 | 70 | 95 | 70 | 4 | 15 | 12 | 145 | 80 |
| **F7349** | 20 | 75 | 100 | 75 | 4 | 15 | 14 | 155 | 100 |
| **16K** | 25 | 85 | 125 | 90 | 4 | 19 | 14 | 175 | 125 |
| | 32 | 95 | 135 | 100 | 4 | 19 | 16 | 175 | 125 |
| | 40 | 100 | 140 | 105 | 4 | 19 | 16 | 190 | 140 |

# SERIES F7350 - JIS 10K
# Cast Steel Angle Hull Valve

Manufactured in accordance to JIS F7400 Flanges as per JIS B2220 - 10K




Sizes 150A to 300A




Sizes 350A to 700A

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Stem** | Stainless Steel | SUS403 |
| **3 - Disc** | Stainless Steel | SUS420J1 |
| **4 - Seat** | Stainless Steel | SUS420J1 |

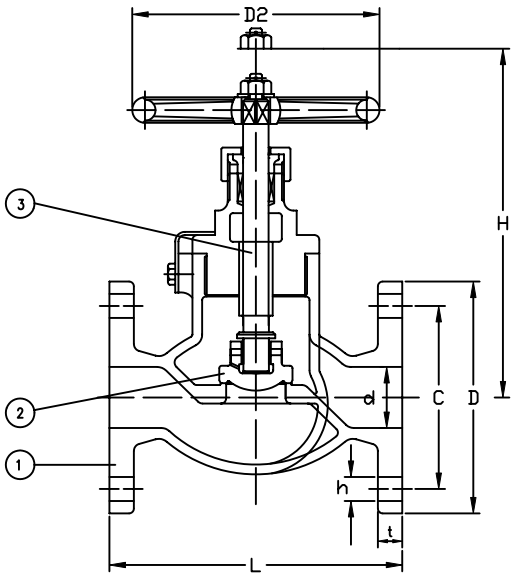
| Nominal Diameter | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| | Body | Seat |
| 200 or less | 1.47 [15] | 1.08 [11] |
| 250 and 300 | 1.03 [10.5] | 0.76 [7.7] |
| 300 more | 0.5 [5.1] | 0.25 [2.5] |

| **Size** | **L1** | **L2** | **D** | **C** | **No.** | **h** | **t** | **H** | **D2** | **Kg** |
|---|---|---|---|---|---|---|---|---|---|---|
| 150 | 205 | - | 280 | 240 | 8 | 23 | 22 | 400 | 315 | 78 |
| 200 | 230 | - | 330 | 290 | 12 | 23 | 22 | 480 | 355 | 117 |
| 250 | 290 | - | 400 | 355 | 12 | 25 | 24 | 535 | 400 | 190 |
| 300 | 320 | - | 445 | 400 | 16 | 25 | 24 | 610 | 450 | 258 |
| 350 | 360 | 320 | 490 | 445 | 16 | 25 | 26 | 1010 | 500 | 376 |
| 400 | 400 | 350 | 560 | 510 | 16 | 27 | 28 | 1130 | 600 | 490 |
| 450 | 450 | 380 | 620 | 565 | 20 | 27 | 30 | 1250 | 710 | 632 |
| 500 | 500 | 430 | 675 | 620 | 20 | 33 | 30 | 1380 | 800 | 839 |
| 550 | 550 | 460 | 745 | 680 | 24 | 33 | 32 | 1505 | 900 | 1060 |
| 600 | 610 | 500 | 795 | 730 | 24 | 33 | 32 | 1630 | 900 | 1270 |
| 650 | 660 | 540 | 845 | 780 | 24 | 33 | 34 | 1765 | 1000 | 1590 |
| 700 | 710 | 570 | 905 | 840 | 24 | 33 | 34 | 1890 | 1000 | 1930 |

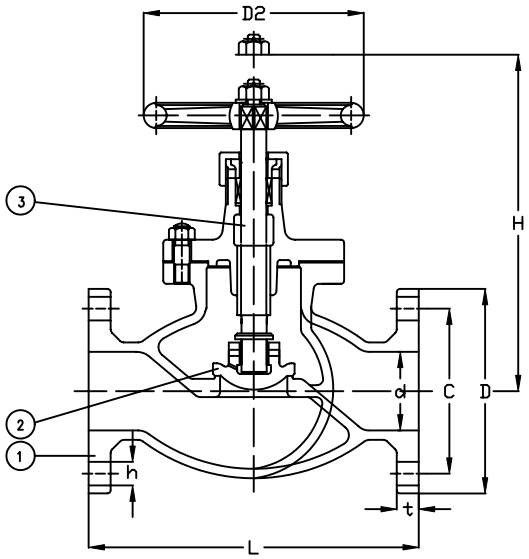
# SERIES F7351 - JIS 5K SERIES F7409 - JIS 16K

## Bronze Globe Valve (SDNR)

Manufactured in accordance to JIS F7400 Flanges as per JIS B2220 - 5K or 16K




Screwed Bonnet
for sizes 15A-40A




Bolted Bonnet
for sizes 50A-65A

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Stem** | Brass | C3771BD |

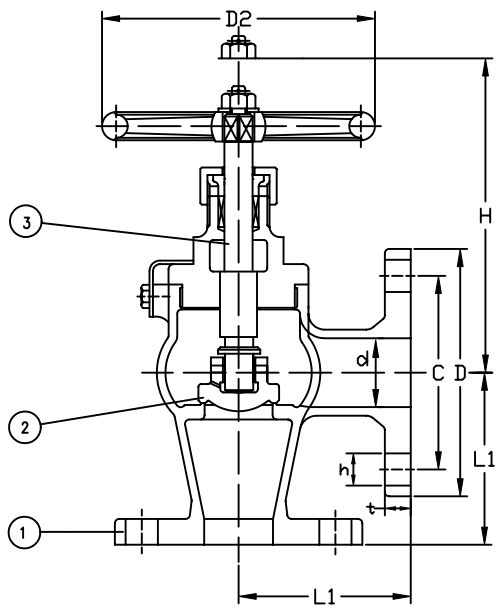
| Nominal Pressure | | 5K | 16K |
|---|---|---|---|
| Hydraulic | Body | 12kg | 38kg |
| Inspection | Seat | 8kg | 23kg |

| Type | Size | L | D | C | No. | h | t | H | D2 |
|---|---|---|---|---|---|---|---|---|---|
| | 15 | 100 | 80 | 60 | 4 | 12 | 9 | 130 | 80 |
| | 20 | 110 | 85 | 65 | 4 | 12 | 10 | 140 | 100 |
| **F7351** | 25 | 120 | 95 | 75 | 4 | 12 | 10 | 160 | 125 |
| **5K** | 32 | 140 | 115 | 90 | 4 | 15 | 12 | 170 | 125 |
| | 40 | 160 | 120 | 95 | 4 | 15 | 12 | 190 | 140 |
| | 50 | 210 | 130 | 105 | 4 | 15 | 14 | 235 | 140 |
| | 65 | 250 | 155 | 130 | 4 | 15 | 14 | 260 | 160 |
| | 15 | 110 | 95 | 70 | 4 | 15 | 12 | 130 | 80 |
| **F7409** | 20 | 120 | 100 | 75 | 4 | 15 | 14 | 140 | 100 |
| **16K** | 25 | 130 | 125 | 90 | 4 | 19 | 14 | 160 | 125 |
| | 32 | 160 | 135 | 100 | 4 | 19 | 16 | 170 | 125 |
| | 40 | 180 | 140 | 105 | 4 | 19 | 16 | 190 | 140 |

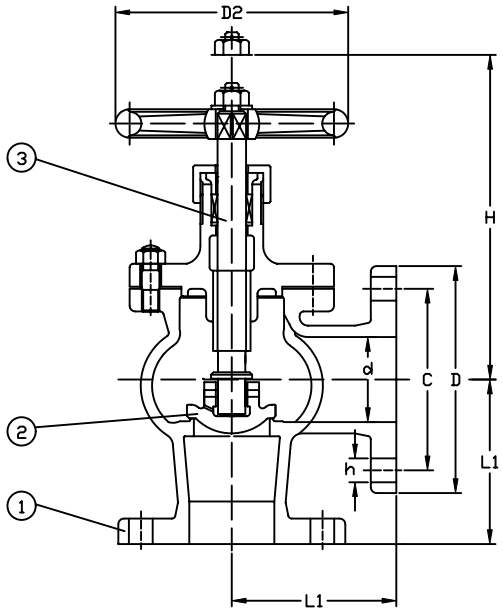
# SERIES F7352 - JIS 5K SERIES F7410 - JIS 16K

## Bronze Angle Valve (SDNR)

Manufactured in accordance to JIS F7400 Flanges as per JIS B2220 - 5K or 16K




Screwed Bonnet
for sizes 15A-40A




Bolted Bonnet
for sizes 50A-65A

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Stem** | Brass | C3771BD |

| Nominal Pressure | | 5K | 16K |
|---|---|---|---|
| Hydraulic | Body | 12kg | 38kg |
| Inspection | Seat | 8kg | 23kg |

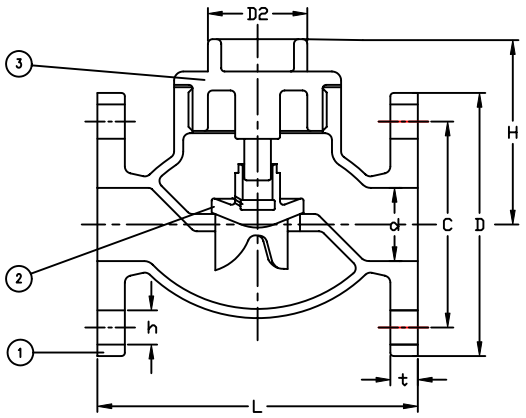
| **Type** | **Size** | **L** | **D** | **C** | **No.** | **h** | **t** | **H** | **D2** |
|---|---|---|---|---|---|---|---|---|---|
| | 15 | 55 | 80 | 60 | 4 | 12 | 9 | 120 | 80 |
| | 20 | 60 | 85 | 65 | 4 | 12 | 10 | 130 | 100 |
| **F7352** | 25 | 65 | 95 | 75 | 4 | 12 | 10 | 145 | 125 |
| **5K** | 32 | 80 | 115 | 90 | 4 | 15 | 12 | 150 | 125 |
| | 40 | 85 | 120 | 95 | 4 | 15 | 12 | 165 | 140 |
| | 50 | 100 | 130 | 105 | 4 | 15 | 14 | 200 | 140 |
| | 65 | 115 | 155 | 130 | 4 | 15 | 14 | 220 | 160 |
| | 15 | 70 | 95 | 70 | 4 | 15 | 12 | 120 | 80 |
| **F7410** | 20 | 75 | 100 | 75 | 4 | 15 | 14 | 130 | 100 |
| **16K** | 25 | 85 | 125 | 90 | 4 | 19 | 14 | 145 | 125 |
| | 32 | 95 | 135 | 100 | 4 | 19 | 16 | 150 | 125 |
| | 40 | 100 | 140 | 105 | 4 | 19 | 16 | 165 | 140 |

# SERIES F7356 - JIS 5K
## Bronze Lift Check Valve

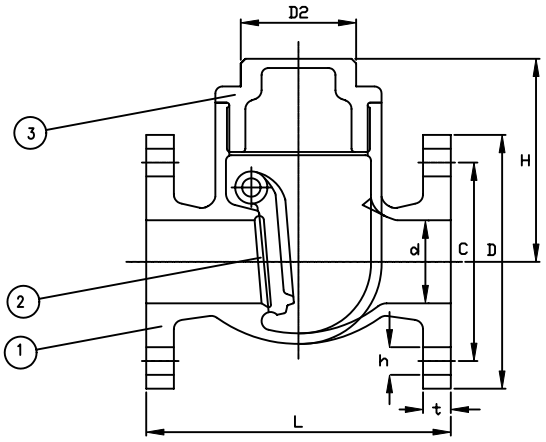
# SERIES F7371- JIS 5K
## Bronze Swing Check Valve

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 5K




Series F7356




Series F7371

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Bonnet** | Bronze | BC6 |

| Nominal Pressure | | 5K |
|---|---|---|
| Hydraulic | Body | 12kg |
| Inspection | Seat | 8kg |

**DIMENSION TABLE FOR JIS F7356**

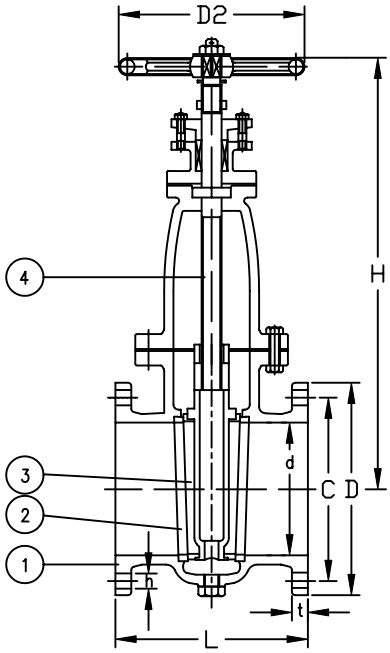
| Size | L | D | C | No. | h | t | H | D2 |
|---|---|---|---|---|---|---|---|---|
| 15 | 100 | 80 | 60 | 4 | 12 | 9 | 72 | 35 |
| 20 | 100 | 85 | 65 | 4 | 12 | 10 | 78 | 37 |
| 25 | 120 | 95 | 75 | 4 | 12 | 10 | 77 | 41 |
| 32 | 140 | 115 | 90 | 4 | 15 | 12 | 81 | 41 |
| 40 | 160 | 120 | 95 | 4 | 15 | 12 | 91 | 46 |

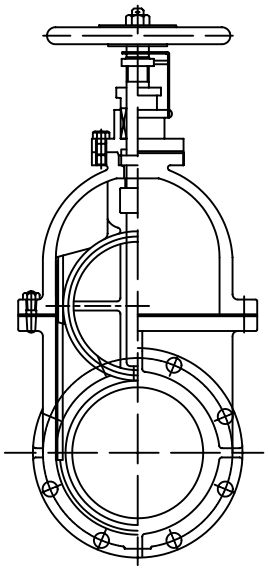
**DIMENSION TABLE FOR JIS F7371**

| Size | L | D | C | No. | h | t | H | D2 |
|---|---|---|---|---|---|---|---|---|
| 15 | 100 | 80 | 60 | 4 | 12 | 9 | 65 | 37 |
| 20 | 110 | 85 | 65 | 4 | 12 | 10 | 68 | 39 |
| 25 | 110 | 95 | 75 | 4 | 12 | 10 | 69 | 41 |
| 32 | 130 | 115 | 90 | 4 | 15 | 12 | 79 | 41 |
| 40 | 140 | 120 | 95 | 4 | 15 | 12 | 93 | 46 |

# SERIES F7363 - JIS 5K
# Cast Iron Gate Valve

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 5K



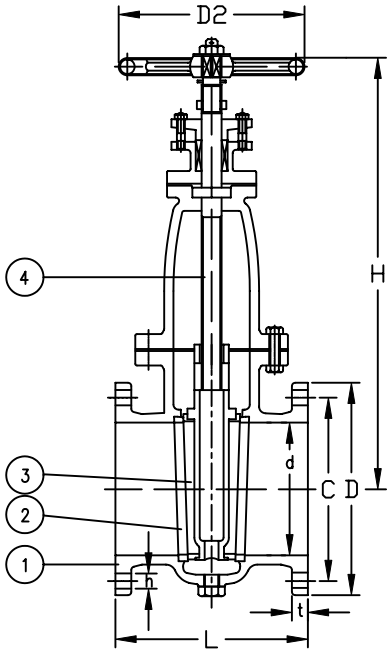
| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Seat** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

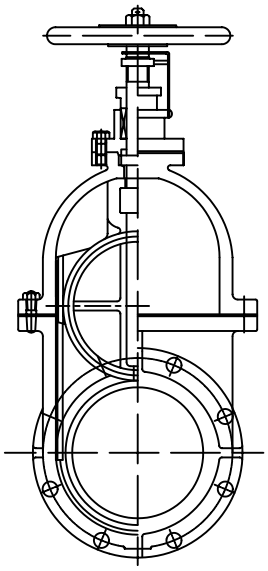
| Nominal Diameter | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| | Body | Seat |
| 400 or less | 1.03 [10.5] | 0.76 [7.7] |
| 450 to 500 | 0.74 [7.5] | 0.54 [5.5] |

| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 180 | 130 | 105 | 4 | 15 | 16 | 285 | 125 | 13.5 |
| 065 | 190 | 155 | 130 | 4 | 15 | 18 | 330 | 140 | 19 |
| 080 | 200 | 180 | 145 | 4 | 19 | 18 | 380 | 160 | 24.2 |
| 100 | 230 | 200 | 165 | 8 | 19 | 20 | 430 | 180 | 35.3 |
| 125 | 250 | 235 | 200 | 8 | 19 | 20 | 495 | 200 | 48.1 |
| 150 | 270 | 265 | 230 | 8 | 19 | 22 | 560 | 224 | 65.3 |
| 200 | 290 | 320 | 280 | 8 | 23 | 24 | 680 | 280 | 103 |
| 250 | 330 | 385 | 345 | 12 | 23 | 26 | 800 | 355 | 164 |
| 300 | 370 | 430 | 390 | 12 | 23 | 28 | 920 | 400 | 236 |
| 350 | 410 | 480 | 435 | 12 | 25 | 30 | 1000 | 450 | 307 |
| 400 | 470 | 540 | 495 | 16 | 25 | 30 | 1100 | 500 | 405 |
| 450 | 500 | 605 | 555 | 16 | 25 | 30 | 1400 | 560 | 581 |
| 500 | 550 | 655 | 605 | 20 | 25 | 32 | 1500 | 560 | 725 |

# SERIES F7363 (S) - JIS 5K
# Cast Steel Gate Valve

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 5K



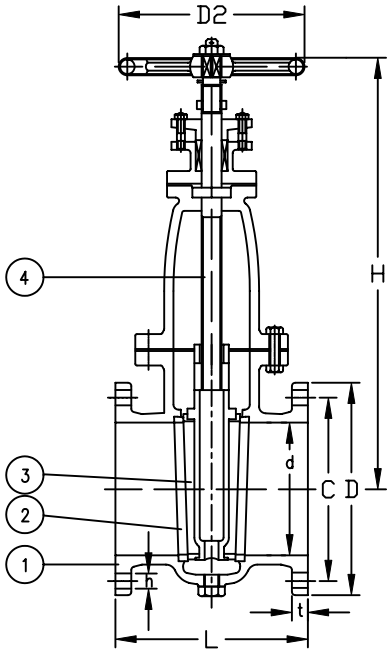
| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Seat** | Stainless Steel | SUS420J1 |
| **3 - Disc** | Stainless Steel | SUS420J1 |
| **4 - Stem** | Stainless Steel | SUS403 |

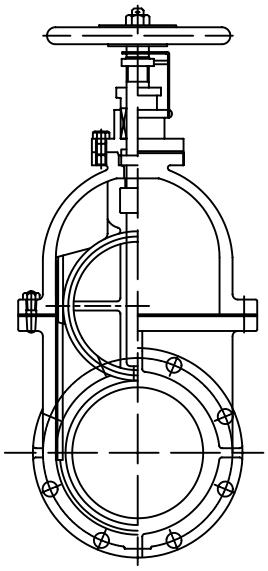
| Nominal Diameter | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| | Body | Seat |
| 50 to 400 | 1.03 [10.5] | 0.76 [7.7] |

| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 180 | 130 | 105 | 4 | 15 | 16 | 285 | 125 | 14.6 |
| 065 | 190 | 155 | 130 | 4 | 15 | 18 | 330 | 140 | 20.6 |
| 080 | 200 | 180 | 145 | 4 | 19 | 18 | 380 | 160 | 26.2 |
| 100 | 230 | 200 | 165 | 8 | 19 | 20 | 430 | 180 | 38.2 |
| 125 | 250 | 235 | 200 | 8 | 19 | 20 | 495 | 200 | 52.1 |
| 150 | 270 | 265 | 230 | 8 | 19 | 22 | 560 | 224 | 70.7 |
| 200 | 290 | 320 | 280 | 8 | 23 | 24 | 680 | 280 | 111.5 |
| 250 | 330 | 385 | 345 | 12 | 23 | 26 | 800 | 355 | 177.6 |
| 300 | 370 | 430 | 390 | 12 | 23 | 28 | 920 | 400 | 255.5 |
| 350 | 410 | 480 | 435 | 12 | 25 | 30 | 1000 | 450 | 382.5 |
| 400 | 470 | 540 | 495 | 16 | 25 | 30 | 1100 | 500 | 438.5 |

# SERIES F7364 - JIS 10K
# Cast Iron Gate Valve

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 10K



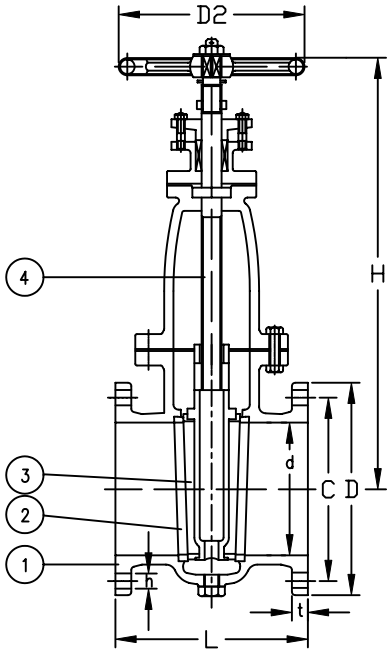
| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Seat** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

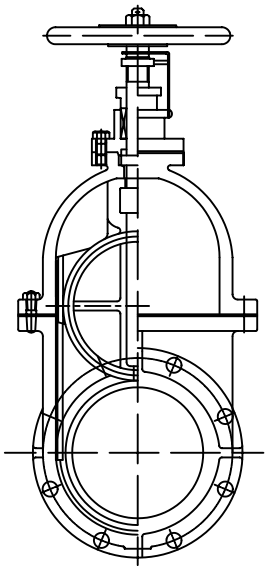
| Nominal Diameter | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| | Body | Seat |
| 400 or less | 2.06 [21] | 1.57 [15.4] |
| 450 to 500 | 1.77 [18] | 1.29 [13.2] |

| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 200 | 155 | 120 | 4 | 19 | 20 | 300 | 140 | 19.4 |
| 065 | 220 | 175 | 140 | 4 | 19 | 22 | 350 | 160 | 27.6 |
| 080 | 230 | 185 | 150 | 8 | 19 | 22 | 400 | 180 | 33 |
| 100 | 250 | 210 | 175 | 8 | 19 | 24 | 450 | 200 | 45.8 |
| 125 | 270 | 250 | 210 | 8 | 23 | 24 | 520 | 224 | 65.9 |
| 150 | 290 | 280 | 240 | 8 | 23 | 26 | 580 | 250 | 86 |
| 200 | 320 | 330 | 290 | 12 | 23 | 26 | 700 | 315 | 136 |
| 250 | 380 | 400 | 355 | 12 | 25 | 30 | 840 | 400 | 230 |
| 300 | 440 | 445 | 400 | 16 | 25 | 32 | 960 | 450 | 331 |
| 350 | 500 | 490 | 445 | 16 | 25 | 34 | 1050 | 500 | 433 |
| 400 | 590 | 560 | 510 | 16 | 27 | 36 | 1150 | 560 | 597 |
| 450 | 640 | 620 | 565 | 20 | 27 | 38 | 1260 | 600 | 704 |
| 500 | 710 | 675 | 620 | 20 | 27 | 40 | 1380 | 680 | 901 |

# SERIES F7366 - JIS 10K
# Cast Steel Gate Valve

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 10K



| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Seat** | Stainless Steel | SUS420J1 |
| **3 - Disc** | Stainless Steel | SUS420J1 |
| **4 - Stem** | Stainless Steel | SUS403 |

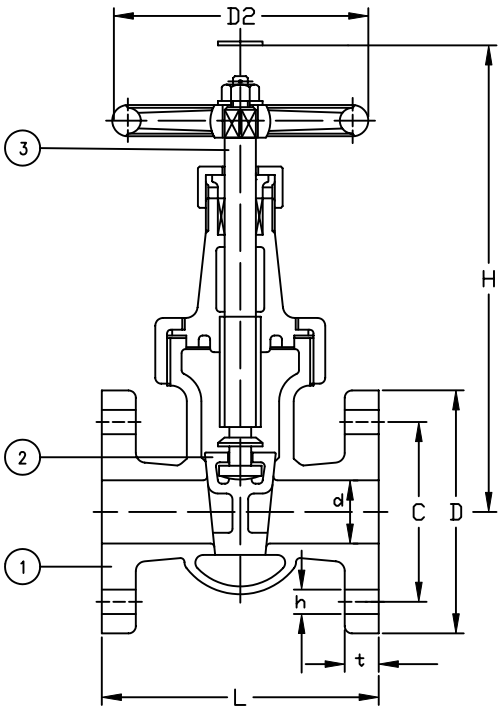
| Nominal Diameter | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| | Body | Seat |
| 50 to 400 | 2.06 [21.0] | 1.51 [15.4] |

| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 200 | 155 | 120 | 4 | 19 | 16 | 300 | 140 | 17.1 |
| 065 | 220 | 175 | 140 | 4 | 19 | 18 | 350 | 160 | 26.5 |
| 080 | 230 | 185 | 150 | 8 | 19 | 18 | 400 | 180 | 31.5 |
| 100 | 250 | 210 | 175 | 8 | 19 | 18 | 450 | 200 | 44.9 |
| 125 | 270 | 250 | 210 | 8 | 23 | 20 | 520 | 224 | 63.2 |
| 150 | 290 | 280 | 240 | 8 | 23 | 22 | 580 | 250 | 82.3 |
| 200 | 310 | 330 | 290 | 12 | 23 | 22 | 700 | 315 | 130 |
| 250 | 340 | 400 | 355 | 12 | 25 | 24 | 840 | 400 | 203 |
| 300 | 380 | 445 | 400 | 16 | 25 | 24 | 960 | 450 | 291 |
| 350 | 420 | 490 | 445 | 16 | 25 | 26 | 1050 | 500 | 368 |
| 400 | 480 | 560 | 510 | 16 | 27 | 28 | 1150 | 560 | 507 |

# SERIES F7367 - JIS 5K SERIES F7368 - JIS 10K

## Bronze Gate Valve

Manufactured in accordance to JIS F7400 Flanges as per JIS B2220 - 5K and 10K



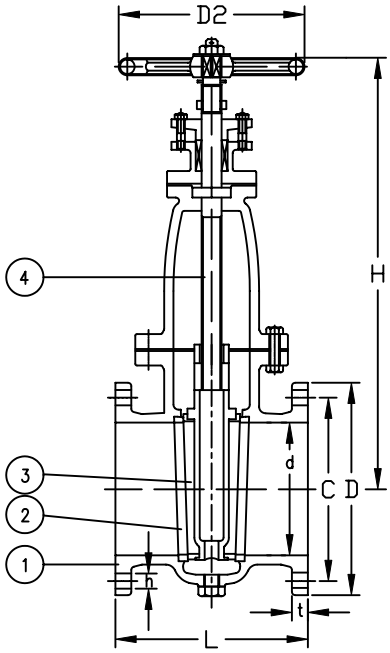

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Stem** | Brass | C3771BD |

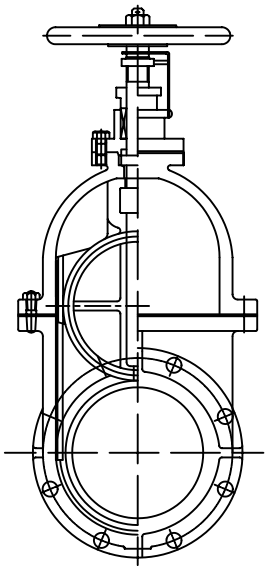
| Nominal Pressure | | 5K | 10K |
|---|---|---|---|
| Hydraulic | Body | 10.5kg | 21kg |
| Inspection | Seat | 7.7kg | 15.4kg |

| Type | Size | L | D | C | No. | h | t | H | D2 |
|---|---|---|---|---|---|---|---|---|---|
| | 15 | 90 | 80 | 60 | 4 | 12 | 9 | 175 | 80 |
| | 20 | 100 | 85 | 65 | 4 | 12 | 10 | 200 | 100 |
| **F7367** | 25 | 110 | 95 | 75 | 4 | 12 | 10 | 220 | 125 |
| **5K** | 32 | 130 | 115 | 90 | 4 | 15 | 12 | 250 | 125 |
| | 40 | 140 | 120 | 95 | 4 | 15 | 12 | 290 | 140 |
| | 15 | 100 | 95 | 70 | 4 | 15 | 12 | 175 | 80 |
| **F7368** | 20 | 110 | 100 | 75 | 4 | 15 | 14 | 200 | 100 |
| **10K** | 25 | 120 | 125 | 90 | 4 | 19 | 14 | 220 | 125 |
| | 32 | 140 | 135 | 100 | 4 | 19 | 16 | 250 | 125 |
| | 40 | 150 | 140 | 105 | 4 | 19 | 16 | 290 | 140 |

# SERIES F7369 - JIS 16K
# Cast Iron Gate Valve

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 16K



| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Seat** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

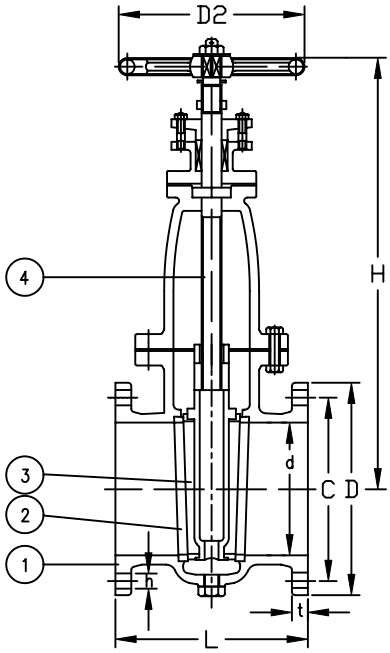
| Nominal Diameter | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| | Body | Seat |
| 50 to 400 | 2.36 [24.0] | 1.73 [17.6] |

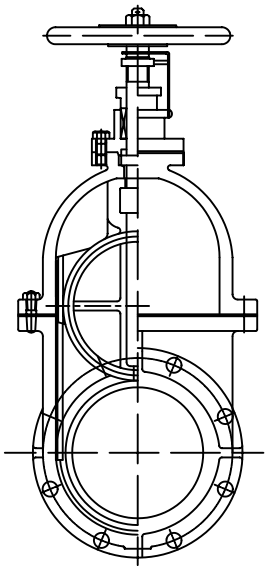
| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 210 | 155 | 120 | 8 | 19 | 20 | 300 | 140 | 19.5 |
| 065 | 240 | 175 | 140 | 8 | 19 | 22 | 350 | 160 | 28 |
| 080 | 260 | 200 | 160 | 8 | 23 | 24 | 400 | 180 | 35 |
| 100 | 290 | 225 | 185 | 8 | 23 | 26 | 450 | 200 | 50 |
| 125 | 310 | 270 | 225 | 8 | 25 | 26 | 520 | 250 | 70 |
| 150 | 330 | 305 | 260 | 12 | 25 | 28 | 580 | 280 | 100 |
| 200 | 370 | 350 | 305 | 12 | 25 | 30 | 700 | 355 | 150 |
| 250 | 430 | 430 | 380 | 12 | 27 | 34 | 845 | 450 | 250 |
| 300 | 490 | 480 | 430 | 16 | 27 | 36 | 1010 | 500 | 423 |
| 350 | 540 | 540 | 480 | 16 | 33 | 38 | 1070 | 560 | 580 |
| 400 | 610 | 605 | 540 | 16 | 33 | 42 | 1160 | 630 | 780 |

# SERIES F7369 - JIS 16K

# Cast Iron Gate Valve - Korean Standard

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 16K



| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Seat** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

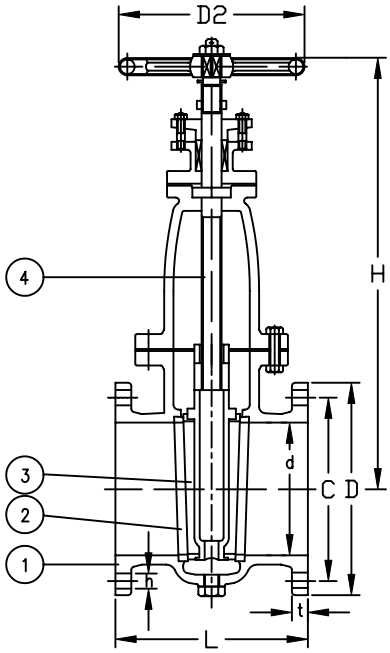
| Nominal Diameter | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| | Body | Seat |
| 50 to 400 | 2.36 [24.0] | 1.73 [17.6] |

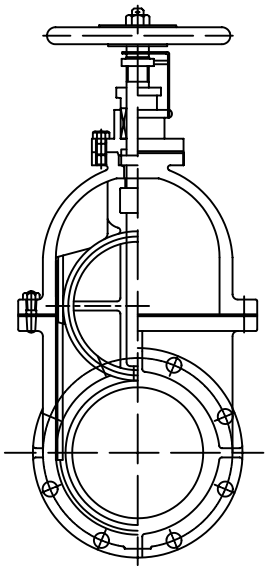
| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 200 | 155 | 120 | 8 | 19 | 16 | 300 | 140 | 22.5 |
| 065 | 220 | 175 | 140 | 8 | 19 | 18 | 350 | 160 | 25 |
| 080 | 230 | 200 | 160 | 8 | 23 | 20 | 400 | 180 | 33 |
| 100 | 250 | 225 | 185 | 8 | 23 | 22 | 450 | 200 | 52 |
| 125 | 270 | 270 | 225 | 8 | 25 | 22 | 520 | 250 | 72 |
| 150 | 290 | 305 | 260 | 12 | 25 | 24 | 580 | 280 | 95 |
| 200 | 310 | 350 | 305 | 12 | 25 | 26 | 700 | 355 | 143 |
| 250 | 350 | 430 | 380 | 12 | 27 | 28 | 845 | 450 | 203 |
| 300 | 490 | 480 | 430 | 16 | 27 | 36 | 1010 | 500 | 430 |
| 350 | 540 | 540 | 480 | 16 | 33 | 38 | 1070 | 560 | 580 |
| 400 | 610 | 605 | 540 | 16 | 33 | 42 | 1160 | 630 | 780 |

# SERIES F7369 - JIS 16K

## Cast Steel Gate Valve - Korean Standard

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 16K



| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Steel | SC49 |
| **2 - Seat** | Stainless Steel | SUS420J1 |
| **3 - Disc** | Stainless Steel | SUS420J1 |
| **4 - Stem** | Stainless Steel | SUS403 |

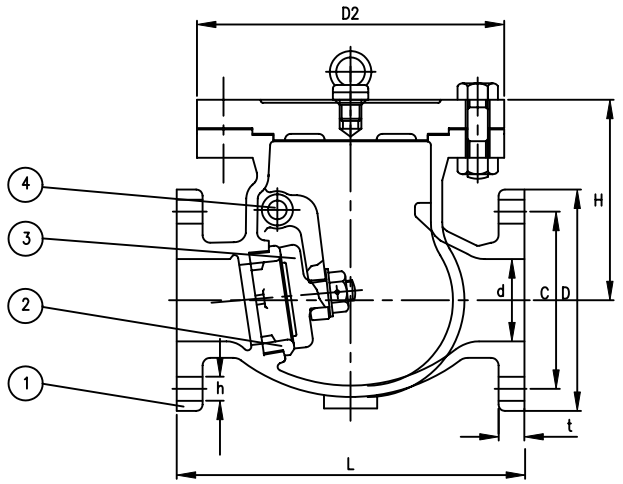
| Nominal Diameter | Test Pressure MPa [kgf/cm2] | |
|---|---|---|
| | Body | Seat |
| 50 to 400 | 2.36 [24.0] | 1.73 [17.6] |

| Size | L | D | C | No. | h | t | H | D2 | Kg |
|---|---|---|---|---|---|---|---|---|---|
| 050 | 200 | 155 | 120 | 8 | 19 | 16 | 300 | 140 | 22.5 |
| 065 | 220 | 175 | 140 | 8 | 19 | 18 | 350 | 160 | 25 |
| 080 | 230 | 200 | 160 | 8 | 23 | 20 | 400 | 180 | 33 |
| 100 | 250 | 225 | 185 | 8 | 23 | 22 | 450 | 200 | 52 |
| 125 | 270 | 270 | 225 | 8 | 25 | 22 | 520 | 250 | 72 |
| 150 | 290 | 305 | 260 | 12 | 25 | 24 | 580 | 280 | 95 |
| 200 | 310 | 350 | 305 | 12 | 25 | 26 | 700 | 355 | 143 |
| 250 | 350 | 430 | 380 | 12 | 27 | 28 | 845 | 450 | 203 |
| 300 | 490 | 480 | 430 | 16 | 27 | 36 | 1010 | 500 | 465 |
| 350 | 540 | 540 | 480 | 16 | 33 | 38 | 1070 | 560 | 628 |
| 400 | 610 | 605 | 540 | 16 | 33 | 42 | 1160 | 630 | 844 |

# SERIES F7372 - JIS 5K SERIES F7373 - JIS 10K

## Cast Iron Swing Check Valves

Manufactured in accordance to JIS F7400 Flanges as per JIS B2220 - 5K & 10K




| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Cast Iron | FC20 |
| **2 - Hinge** | Brass | C3771BD |
| **3 - Disc** | Bronze | BC6 |
| **4 - Seat** | Bronze | BC6 |

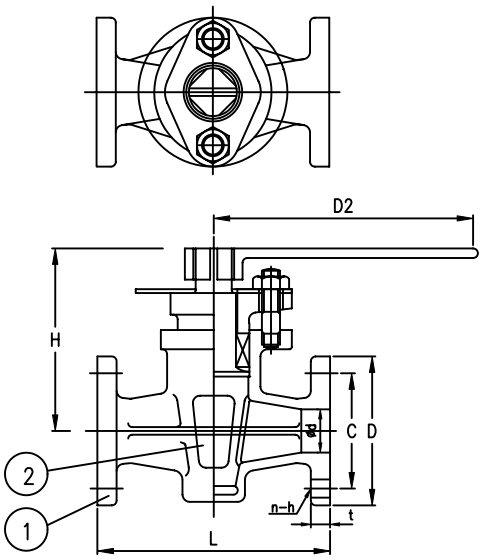
| Nominal Pressure | | 5K | 16K |
|---|---|---|---|
| Hydraulic | Body | 1.03 [10.5] | 2.06 [21.0] |
| Inspection | Seat | 0.76 [7.7] | 1.51 [15.4] |

| Type | Size | L | D | C | No. | h | t | H | D2 | KG |
|---|---|---|---|---|---|---|---|---|---|---|
| | 050 | 190 | 155 | 120 | 4 | 19 | 16 | 97 | 135 | 9.06 |
| | 065 | 220 | 175 | 140 | 4 | 19 | 18 | 119 | 155 | 13.9 |
| | 080 | 250 | 185 | 150 | 8 | 19 | 18 | 129 | 185 | 19.1 |
| **F7372** | 100 | 280 | 210 | 175 | 8 | 19 | 18 | 146 | 205 | 27.1 |
| **5K** | 125 | 330 | 250 | 210 | 8 | 23 | 20 | 171 | 235 | 39 |
| | 150 | 380 | 280 | 240 | 8 | 23 | 22 | 198 | 265 | 53.7 |
| | 200 | 460 | 330 | 290 | 12 | 23 | 22 | 235 | 300 | 94.4 |
| | 250 | 550 | 400 | 355 | 12 | 25 | 24 | 290 | 390 | 150 |
| | 050 | 210 | 155 | 120 | 4 | 19 | 20 | 109 | 150 | 13.6 |
| | 065 | 240 | 175 | 140 | 4 | 19 | 22 | 126 | 170 | 19.6 |
| **F7373** | 080 | 270 | 185 | 150 | 8 | 19 | 22 | 136 | 185 | 22.6 |
| **10K** | 100 | 300 | 210 | 175 | 8 | 19 | 24 | 153 | 205 | 30.9 |
| | 125 | 350 | 250 | 210 | 8 | 23 | 24 | 180 | 250 | 49.2 |
| | 150 | 400 | 280 | 240 | 8 | 23 | 26 | 205 | 275 | 64.9 |
| | 200 | 480 | 330 | 290 | 12 | 23 | 26 | 242 | 330 | 103 |

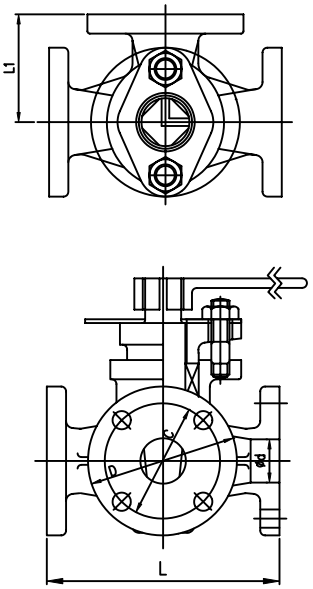
# SERIES F7381 - JIS 5K
# Bronze Cocks - 2 & 3 Way

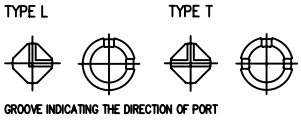
Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 5K

Two-Way Cock




Three-Way Cock




| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Plug** | Bronze | BC6 |

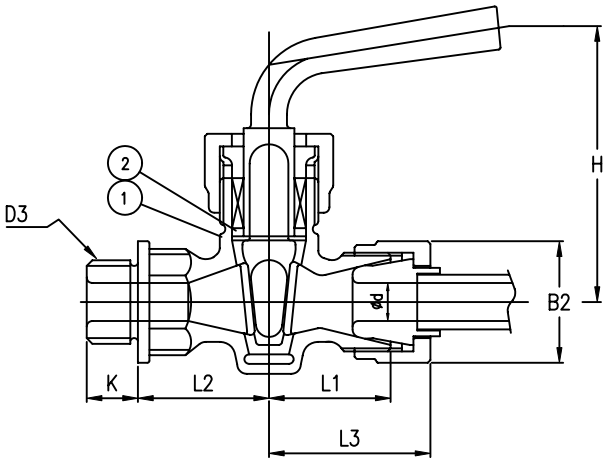
| | | |
|---|---|---|
| Hydraulic | Body | 7.5kg |
| Inspection | Plug | 5.5kg |

| **Size** | **L** | **L1** | **D** | **C** | **No.** | **h** | **t** | **H** | **D2** |
|---|---|---|---|---|---|---|---|---|---|
| 25 | 130 | 65 | 95 | 75 | 4 | 12 | 10 | 105 | 160 |
| 32 | 160 | 80 | 115 | 90 | 4 | 15 | 12 | 125 | 180 |
| 40 | 170 | 85 | 120 | 95 | 4 | 15 | 12 | 135 | 190 |
| 50 | 190 | 95 | 130 | 105 | 4 | 15 | 14 | 163 | 200 |
| 65 | 220 | 110 | 155 | 130 | 4 | 15 | 14 | 201 | 250 |

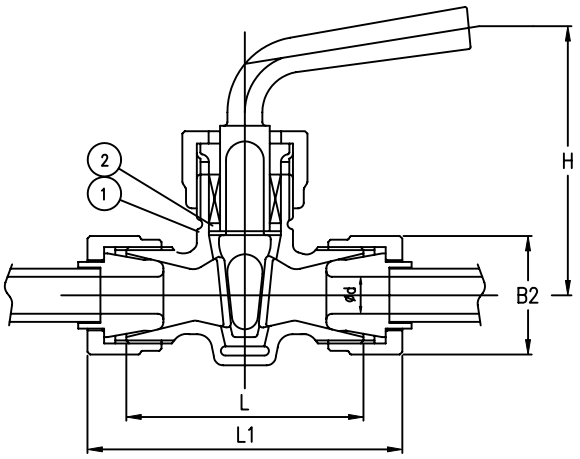
# SERIES F7387 - JIS 16K

# Bronze Cock - S & U Types

Manufactured in accordance to JIS F7400




"S" Type




"U" Type

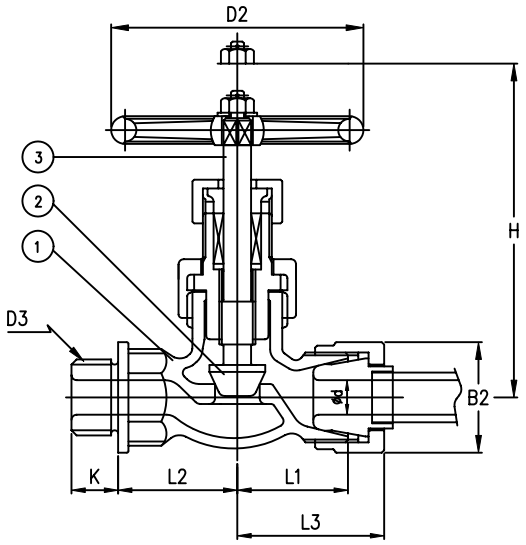
| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Plug** | Bronze | BC6 |

| | Test Pressure | |
|---|---|---|
| Nominal Diam | Body | Plug |
| 6 and 10 | 30kg | 22kg |
| 15 and 20 | 24kg | 17.6kg |

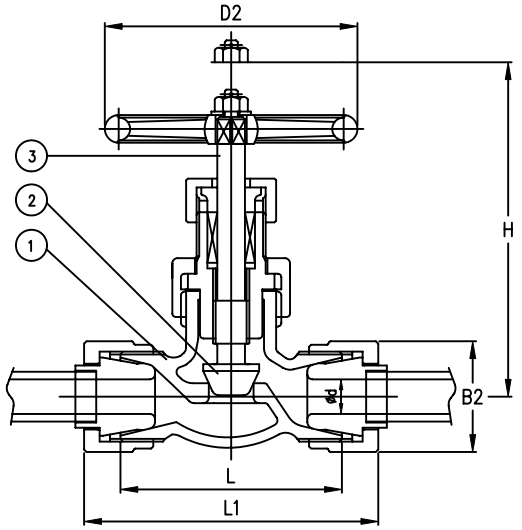
| **Type** | **Size** | **Pipe Out Dia** | | **L** | **L1** | **L2** | **L3** | **K** | **d3** | **B2** | **H** |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **Copper** | **Steel** | | | | | | | | |
| S Type | 6 | 8 / 10 | 10.5 | - | 28 | 28 | 37 | 12 | M20x1.5 | 30 | 65 |
| | 10 | 15 | 17.3 | - | 40 | 35 | 52 | 16 | M24x2 | 41 | 73 |
| | 15 | 20 | 21.7 | - | 45 | 45 | 58 | 22 | M30x2 | 46 | 92 |
| | 20 | 25 | 27.2 | - | 50 | 50 | 65 | 24 | M36x2 | 50 | 104 |
| U Type | 6 | 8 / 10 | 10.5 | 56 | 74 | - | - | - | - | 30 | 65 |
| | 10 | 15 | 17.3 | 80 | 104 | - | - | - | - | 41 | 73 |

# SERIES F7388 - JIS 20K
# Bronze Globe - S & U Types

Manufactured in accordance to JIS F7400




"S" Type




"U" Type

| **Part Nr.** | **Material** | **Code** |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Stem** | Brass | C3771BD |
| **3 - Disc** | Bronze | BC6 |

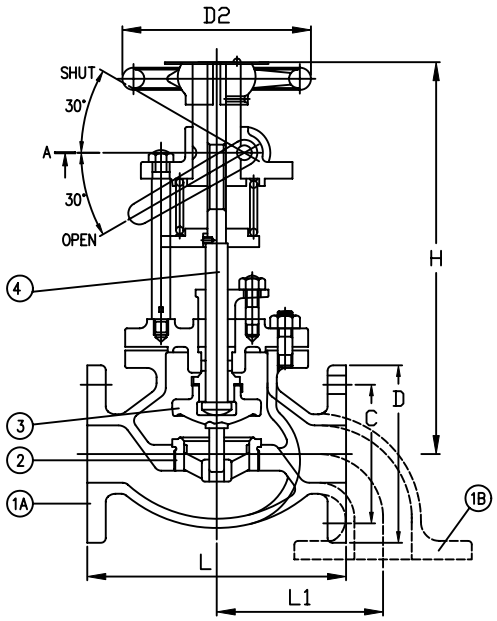
| | Test Pressure | |
|---|---|---|
| Nominal Diam | Body | Plug |
| 6 and 10 | 42kg | 30.8kg |

| **Type** | **Size** | **Pipe Out Dia** | | **L** | **L1** | **L2** | **L3** | **K** | **d3** | **B2** | **H** | **D2** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **Copper** | **Steel** | | | | | | | | | |
| S Type | 6 | 8 / 10 | 10.5 | - | 32 | 32 | 41 | 12 | M20x1.5 | 30 | 100 | 63 |
| | 10 | 15 | 17.3 | - | 36 | 35 | 48 | 16 | M24x2 | 41 | 115 | 80 |
| U Type | 6 | 8 / 10 | 10.5 | 64 | 82 | - | - | - | - | 30 | 100 | 63 |
| | 10 | 15 | 17.3 | 72 | 96 | - | - | - | - | 41 | 115 | 80 |

# SERIES F7399 - JIS 5K and10K

# Emergency Shut-Off Valve - Types “A” and “B”

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 5K or 10K




| | | |
|---|---|---|
| Hydraulic | Body | 7kg |
| Inspection | Seat | 2kg |

| For Sizes 25A & 40A | | |
|---|---|---|
| **Part Nr.** | **Mat** | **Code** |
| **1A - Body** | Bronze | BC6 |
| **3 - Disc** | Bronze | BC6 |
| **4 - Stem** | Brass | C3771BD |

| For Sizes 50A & Higher | | |
|---|---|---|
| **Part Nr.** | **Mat** | **Cod** |
| **1B - Body** | Cast Steel | SC49 |
| **2 - Seat** | Stainless Steel | SCS2 |
| **3 - Disc** | Stainless Steel | SCS2 |
| **4 - Stem** | Stainless Steel | SUS403 |

| | 5K Flanged | | | | | | | | 10K Flanged | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Size** | **L** | **L1** | **L2** | **D** | **C** | **No.** | **h** | **t** | **L** | **L1** | **L2** | **D** | **C** | **No.** | **h** | **t** | **H** | **D1** |
| 25 | 130 | 75 | 60 | 95 | 75 | 4 | 12 | 10 | 130 | 80 | 65 | 125 | 90 | 4 | 19 | 14 | 265 | 120 |
| 40 | 160 | 100 | 75 | 120 | 95 | 4 | 15 | 12 | 170 | 85 | 80 | 140 | 105 | 4 | 19 | 16 | 310 | 140 |
| 50 | 210 | 125 | 90 | 130 | 105 | 4 | 15 | 14 | 220 | 125 | 90 | 155 | 120 | 4 | 19 | 16 | 335 | 160 |
| 65 | 250 | 155 | 100 | 155 | 130 | 4 | 15 | 14 | 270 | 155 | 100 | 175 | 140 | 4 | 19 | 18 | 395 | 180 |
| 80 | 280 | 165 | 120 | 180 | 145 | 4 | 19 | 18 | 300 | 165 | 120 | 185 | 150 | 8 | 19 | 18 | 420 | 180 |
| 100 | 340 | 200 | 130 | 200 | 165 | 8 | 19 | 18 | 350 | 200 | 130 | 210 | 175 | 8 | 19 | 18 | 510 | 224 |
| 125 | 410 | 250 | 160 | 235 | 200 | 8 | 19 | 20 | 420 | 250 | 160 | 250 | 210 | 8 | 19 | 20 | 440 | 250 |
| 150 | 480 | 300 | 180 | 265 | 230 | 8 | 19 | 22 | 490 | 300 | 180 | 280 | 240 | 8 | 19 | 22 | 520 | 320 |

# SERIES F7415 - JIS 5K
Bronze Globe Lift Check Valve

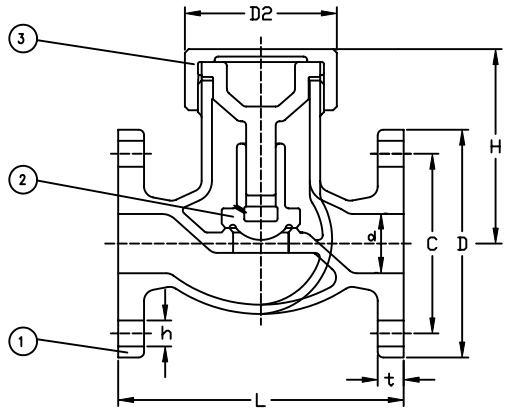
# SERIES F7416 - JIS 5K
Bronze Angle Lift Check Valve

# SERIES F7417 - JIS 16K
Bronze Globe Lift Check Valve

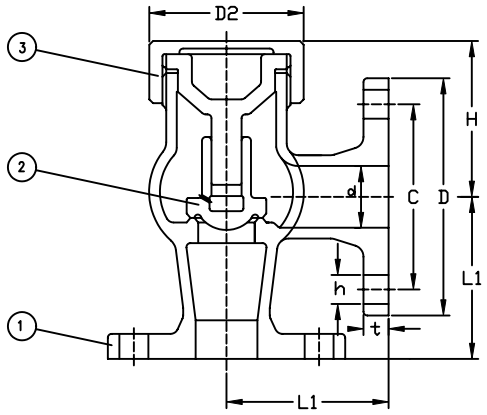
# SERIES F7418 - JIS 16K
Bronze Angle Lift Check Valve

Manufactured in accordance to JIS F7400

Flanges as per JIS B2220 - 5K & 16K




Series F7415 & F7417




Series F7416 & F7418

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Bronze | BC6 |
| **2 - Disc** | Bronze | BC6 |
| **3 - Bonnet** | Bronze | BC6 |

| Nominal Pressure | | 5K | 16K |
|---|---|---|---|
| Hydraulic | Body | 12kg | 38kg |
| Inspection | Seat | 8kg | 23kg |

**DIMENSION TABLE FOR JIS F7415 AND F7416**

| Size | L | L1 | D | C | No. | h | t | H | H1 | D2 |
|---|---|---|---|---|---|---|---|---|---|---|
| 15 | 100 | 55 | 80 | 60 | 4 | 12 | 9 | 66 | 56 | 55 |
| 20 | 110 | 60 | 85 | 65 | 4 | 12 | 10 | 71 | 59 | 55 |
| 25 | 120 | 65 | 95 | 75 | 4 | 12 | 10 | 81 | 67 | 60 |
| 32 | 140 | 80 | 115 | 90 | 4 | 15 | 12 | 83 | 65 | 70 |
| 40 | 160 | 85 | 120 | 95 | 4 | 15 | 12 | 91 | 69 | 80 |

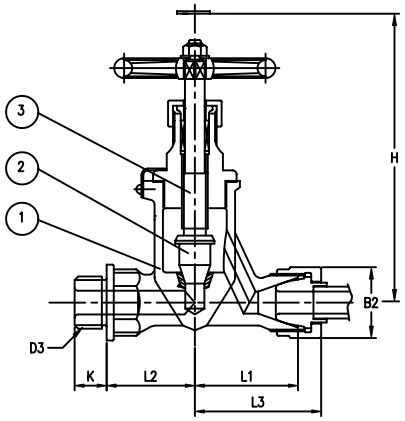
**DIMENSION TABLE FOR JIS F7417 AND F7418**

| Size | L | L1 | D | C | No. | h | t | H | H1 | D2 |
|---|---|---|---|---|---|---|---|---|---|---|
| 15 | 110 | 70 | 95 | 70 | 4 | 15 | 12 | 66 | 56 | 55 |
| 20 | 120 | 75 | 100 | 75 | 4 | 15 | 14 | 71 | 59 | 55 |
| 25 | 130 | 85 | 125 | 90 | 4 | 19 | 14 | 81 | 67 | 60 |
| 32 | 160 | 95 | 135 | 100 | 4 | 19 | 16 | 83 | 65 | 70 |
| 40 | 180 | 100 | 140 | 105 | 4 | 19 | 16 | 91 | 69 | 80 |

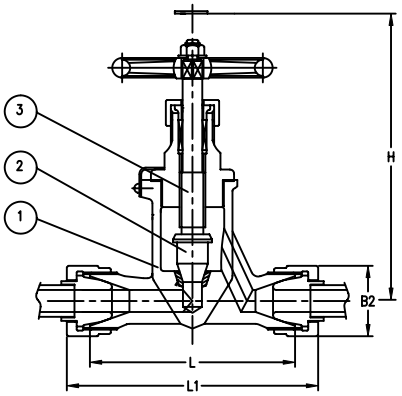
# SERIES F7421 - JIS 20K

# Forged Steel Globe Valve - Type “S”, “U” and “F”

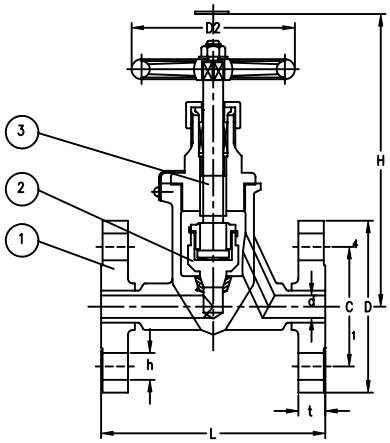
Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 20K




“S” Type




“U” Type




“F” Type

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Forged Steel | SF45 |
| **2 - Stem** | Stainless Steel | SUS403 |
| **3 - Disc** | Stainless Steel | SUS402J1 |

| | Test Pressure MPa [kg/cm2] |
|---|---|
| Body | 5.0 [51.0] |
| Seat | 3.67 [37.4] |

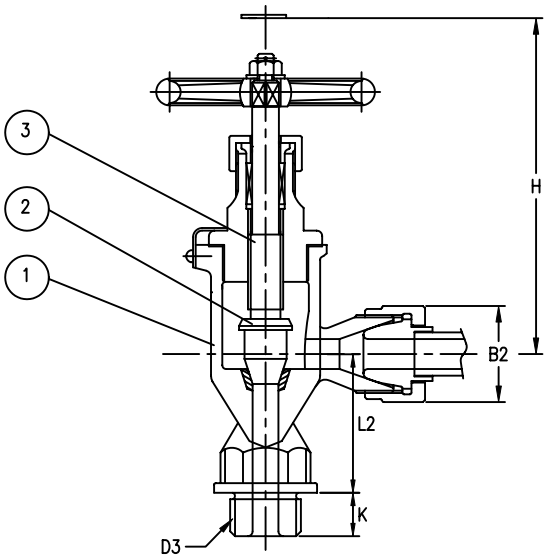
| Type | Type of Connection | Nominal Dia. |
|---|---|---|
| S | Screw thread type | 6 and 10 |
| U | Union type | 6 and 10 |
| F | Flange type | 15 to 25 |

| | Size | L | L1 | L2 | L3 | K | D | C | n | h | t | H | B2 | D3 | Kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Type | 6 | - | 42 | 38 | 51 | 12 | - | - | - | - | - | 125 | 30 | M20x1.5 | 1.23 |
| “S” | 10 | - | 51 | 45 | 63 | 16 | - | - | - | - | - | 140 | 41 | M42x2 | 1.59 |
| Type | 6 | 84 | 102 | - | - | - | - | - | - | - | - | 125 | 30 | - | 1.26 |
| “U” | 10 | 102 | 126 | - | - | - | - | - | - | - | - | 140 | 41 | - | 1.72 |
| Type | 15 | 140 | - | - | - | - | 95 | 70 | 4 | 15 | 14 | 180 | - | - | 3.90 |
| “F” | 20 | 160 | - | - | - | - | 100 | 75 | 4 | 15 | 16 | 205 | - | - | 5.29 |
| | 25 | 180 | - | - | - | - | 125 | 90 | 4 | 19 | 16 | 225 | - | - | 7.49 |

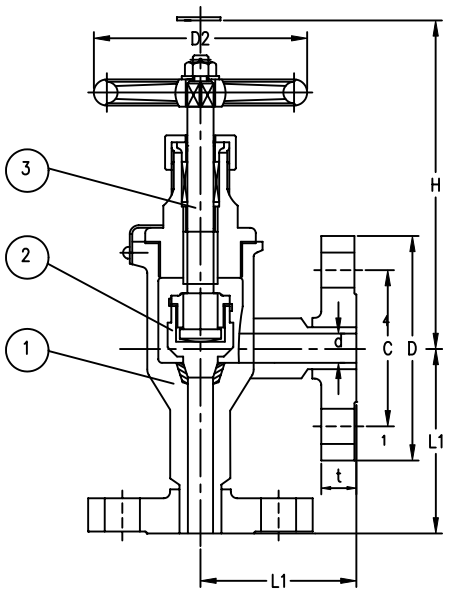
# SERIES F7422 - JIS 20K

## Forged Steel Angle Valve - Type “S” and “F”

Manufactured in accordance to JIS F7400 — Flanges as per JIS B2220 - 20K




“S” Type




“F” Type

| Part Nr. | Material | Code |
|---|---|---|
| **1 - Body** | Forged Steel | SF45 |
| **2 - Stem** | Stainless Steel | SUS403 |
| **3 - Disc** | Stainless Steel | SUS402J1 |

| | Test Pressure MPa [kg/cm2] |
|---|---|
| Body | 5.0 [51.0] |
| Seat | 3.76 [37.4] |

| Type | Type of Connection | Nominal Dia. |
|---|---|---|
| S | Screw thread type | 6 and 10 |
| F | Flange type | 15 to 25 |

| | Size | L | L1 | L2 | L3 | K | D | C | n | h | t | H | B2 | D3 | Kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Type | 6 | - | 42 | 38 | 51 | 12 | - | - | - | - | - | 115 | 30 | M20x1.5 | 1.20 |
| “S” | 10 | - | 46 | 45 | 58 | 16 | - | - | - | - | - | 125 | 41 | M24x2 | 1.54 |
| Type | 15 | 75 | - | - | - | - | 95 | 70 | 4 | 15 | 14 | 160 | - | - | 3.8 |
| “F” | 20 | 80 | - | - | - | - | 100 | 75 | 4 | 15 | 16 | 180 | - | - | 5.05 |
| | 25 | 95 | - | - | - | - | 125 | 90 | 4 | 19 | 16 | 190 | - | - | 7.16 |

# Available Materials for Butterfly Valve Series: 50, 52, 53, 54, 55, 56 and 58

| Part | Material | DIN Code | ASTM Code |
|---|---|---|---|
| Body: | Cast Iron | GG 25 | ASTM A126 Cl. B |
| | Ductile Iron | GGG 40 | ASTM A536 Gr 65-45-12 |
| | Carbon Steel | GS-C 25 | ASTM A216 WCB |
| | Stainless Steel | G-X6 CrNiMo 18.10 | ASTM A351 Gr CF8-CF8M |
| | Bronze | RG5 | ASTM B62 |
| Disc: | Ductile Iron | GGG 40 | ASTM A536 Gr 65-45-12 |
| | Stainless Steel | G-X6 CrNiMo 18.10 | ASTM A351 Gr CF8-CF8M |
| | Aluminum Bronze | G-CuAl10Ni | ASTM B 148 Cl. C 95500 |
| Stem: | Stainless Steel | X2 CrNiMo 18.10 | ASTM A276 316 |
| Seat: | NBR | Nitrile Rubber | Temp Range -25° C / + 100° C |
| | EPDM | Copolymer EPDM | Temp Range -35° C / + 130° C |
| | Viton | Fluoelastomer | Temp Range -20° C / + 200° C |
| | Teflon | PTFE | Temp Range -80° C / + 232° C |
| Packing: | NBR | Nitrile Rubber | Temp Range -25° C / + 100° C |
| | EPDM | Copolymer EPDM | Temp Range -35° C / + 130° C |
| | Viton | Fluoelastomer | Temp Range -20° C / + 200° C |

# Butterfly Valves
## Material Selection

| Chemical Agent | Bronze | Cast Iron | Cast Steel | SS316 | Monel | EPDM | NBR | VITON | Narual Rubber |
|---|---|---|---|---|---|---|---|---|---|
| Acetone | E | G | G | E | E | G | U | U | E |
| Acetylene | E | G | G | E | - | U | F | U | - |
| Alcohol methyl | E | U | - | E | E | E | G | U | G |
| Alumina | G | G | - | G | G | E | E | G | G |
| Amonia solutions | U | F | - | E | G | E | G | U | G |
| Benzene | E | F | G | E | G | U | U | G | U |
| Bunker oil | F | U | U | E | E | U | F | F | U |
| Carbon dioxide | E | F | G | E | - | G | G | E | G |
| Chlorobenzene | E | F | F | E | E | U | U | G | U |
| Crude oil | U | U | G | E | E | U | F | E | U |
| Diesel fuel | F | U | E | E | E | U | E | E | U |
| Ethyl glycol | E | G | G | E | E | E | E | E | G |
| Fruit juices | U | U | - | E | - | G | G | - | - |
| Fuel oil | E | F | E | E | G | U | F | E | U |
| Gasoline | G | F | G | E | G | U | E | E | U |
| Glycerine glycerol | G | F | E | E | G | - | E | E | - |
| Hydraulic oil (petroleum based) | U | U | E | F | F | U | F | F | F |
| Hydraulic oil (water based) | E | U | E | E | U | E | F | U | F |
| Kerosene | E | E | E | E | E | U | E | E | U |
| Lubricating oil | E | E | E | E | E | U | E | E | U |
| Milk | - | U | E | E | - | E | E | E | E |
| Mineral oil | - | F | E | F | - | U | E | E | U |
| Napthalene | E | F | E | E | - | U | U | G | U |

| Chemical Agent | Bronze | Cast Iron | Cast Steel | SS316 | Monel | EPDM | NBR | VITON | Narual Rubber |
|---|---|---|---|---|---|---|---|---|---|
| Natural gas | E | G | E | E | - | U | E | E | U |
| Olive oil | - | - | E | E | - | G | E | E | U |
| Oxygen 200 F | E | E | E | E | E | E | G | E | F |
| Oxygen 300 F | E | E | E | E | E | U | U | G | U |
| Petroleum | F | U | E | G | G | U | E | E | U |
| Phenol | - | U | E | E | E | U | U | E | U |
| Potassium nitrate | F | F | E | E | G | E | E | E | E |
| Potassium phosphate | - | U | - | G | - | - | E | E | - |
| Propane | E | F | E | E | - | U | E | E | - |
| Resins | E | U | - | E | - | - | - | G | F |
| Sea Water | G | F | F | G | E | E | G | E | - |
| Sodium chloride | E | U | G | E | E | E | E | E | E |
| Sodium nitrate | G | U | G | E | G | E | G | - | G |
| Steam 300 F | G | U | E | E | E | E | U | U | U |
| Sulphur molten | U | U | F | G | U | E | - | E | - |
| Sulphur dioxite | F | U | E | E | E | G | U | E | U |
| Sulphur trioxide | - | U | G | E | - | - | U | E | U |
| Tar | E | F | E | E | E | U | U | G | U |
| Turpentine | G | U | G | E | | U | U | E | U |
| Varnish | E | E | E | E | E | U | F | E | U |
| Water distiled | U | U | G | E | - | E | G | - | - |
| Water fresh | E | F | G | E | E | E | G | - | E |
| Water sewage | E | U | G | E | E | E | G | - | - |

(E) - Excellent

(G) - Good

(F) - Fair

(U) - Unsatisfactory

(-) - Insufficient Data Available

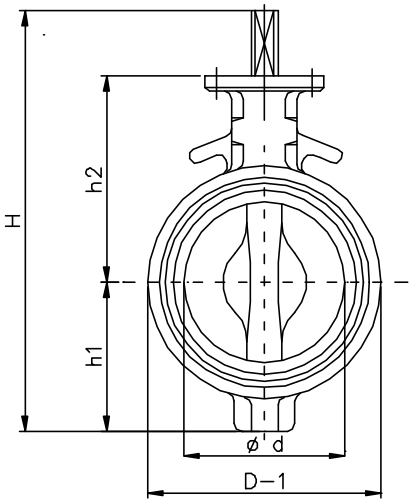
The information stated above is intended as a recommendation and suggestion based on current available data. World Wide Metric, Inc. assumes no liability or responsibility for results or damages incurred through the application of the data presented.

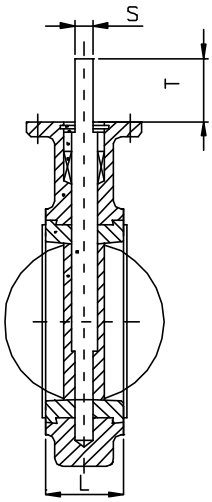
WORLD WIDE METRIC

# Butterfly Valve, Wafer, Lug & Flanged

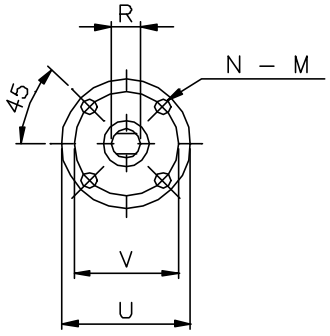
**Series ISO 5752 S - 20**

Wafer type.....Series 50

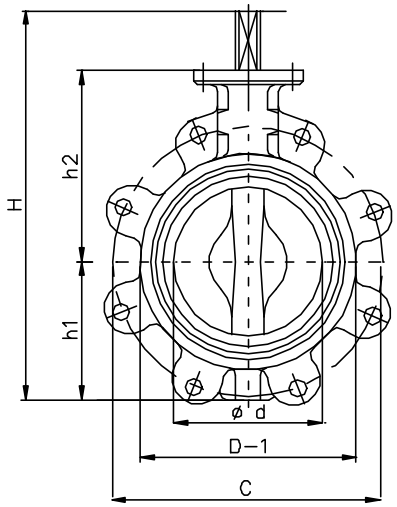


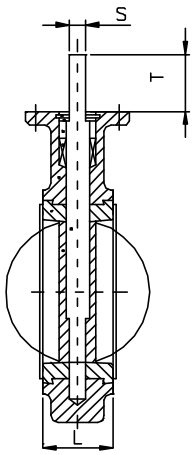




Top Flange to ISO5711

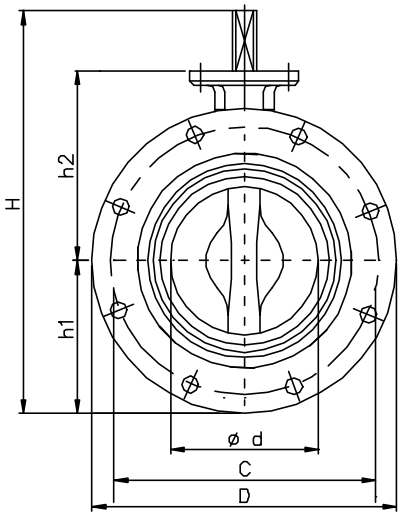


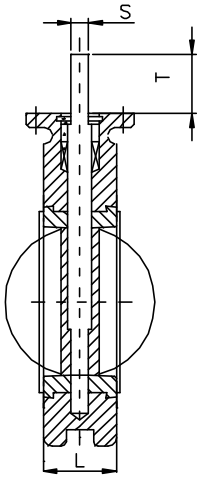

Lug type.....Series 52




Flanged type...Series 55



WORLD WIDE METRIC

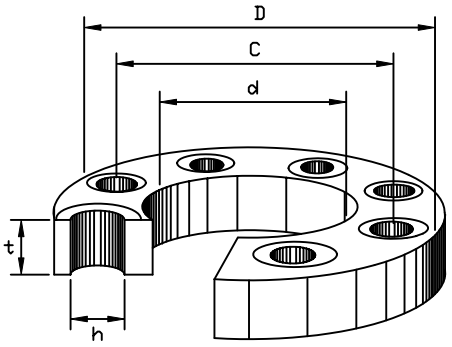
# Butterfly Valve—Wafer, Lug & Flanged
## Series ISO 5752 - S20

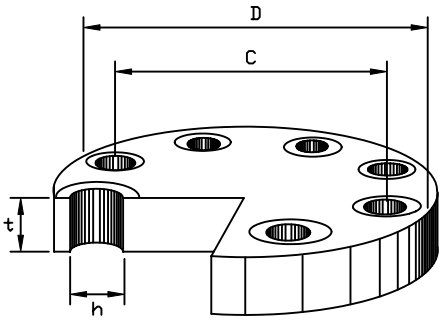
| Size | | | | | | | Stem | | | Top flange to ISO 5211 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | D-1 | d | L | H | h1 | h2 | R | S | T | U | V | N | M |
| 40 | 86 | 40 | 33 | 205 | 58 | 124 | 10 | 8 | - | 90 | 70 | 4 | 9 |
| 50 | 102 | 52 | 43 | 240 | 78 | 130 | 14 | 10 | - | 90 | 70 | 4 | 9 |
| 65 | 115 | 65 | 46 | 247 | 88 | 130 | 14 | 10 | - | 90 | 70 | 4 | 9 |
| 80 | 128 | 80 | 46 | 266 | 93 | 143 | 14 | 10 | - | 90 | 70 | 4 | 9 |
| 100 | 153 | 100 | 52 | 298 | 105 | 161 | 16 | 10 | - | 90 | 70 | 4 | 9 |
| 125 | 183 | 125 | 56 | 331 | 125 | 177 | 19 | 12 | - | 90 | 70 | 4 | 9 |
| 150 | 210 | 150 | 56 | 363 | 140 | 191 | 19 | 15 | - | 90 | 70 | 4 | 9 |
| 200 | 259 | 198 | 60 | 412 | 163 | 217 | 22 | 15 | - | 90 | 70 | 4 | 9 |
| 250 | 328 | 248 | 68 | 506 | 205 | 267 | 28 | 18 | - | 125 | 102 | 4 | 12 |
| 300 | 374 | 298 | 78 | 538 | 249 | 295 | 28 | 20 | - | 125 | 102 | 4 | 12 |
| 350 | 416 | 327 | 78 | 675 | 271 | 324 | 32 | 20 | 45 | 125 | 102 | 4 | 12 |
| 400 | 475 | 387 | 102 | 817 | 343 | 369 | 35 | - | 60 | 175 | 140 | 4 | 18 |
| 450 | 535 | 438 | 114 | 877 | 366 | 406 | 38 | - | 60 | 175 | 140 | 4 | 18 |
| 500 | 586 | 477 | 127 | 1008 | 421 | 485 | 45 | - | 60 | 210 | 165 | 4 | 22 |
| 600 | 695 | 560 | 154 | 1180 | 493 | 570 | 50 | - | 60 | 210 | 165 | 4 | 22 |
| 700 | 800 | 690 | 165 | 1355 | 580 | 574 | 70 | - | 120 | 300 | 254 | 8 | 18 |
| 750 | 857 | 729 | 190 | 1445 | 656 | 615 | 70 | - | 120 | 300 | 254 | 8 | 18 |
| 800 | 920 | 785 | 190 | 1495 | 630 | 669 | 80 | - | 140 | 300 | 254 | 8 | 18 |
| 900 | 1020 | 900 | 203 | 1625 | 690 | 671 | 80 | - | 140 | 350 | 298 | 8 | 22 |
| 1000 | 1125 | 990 | 216 | 1802 | 790 | 842 | 90 | - | 140 | 350 | 298 | 8 | 22 |

| Size | Weight | | | JIS 5K | | | JIS 10K | | | JIS 16K | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Wafer | Lug | Flanged | D | C | Bolt | D | C | Bolt | D | C | Bolt |
| 40 | 3.1 | 3.5 | 6.2 | 120 | 95 | 4x15 | 140 | 105 | 4x19 | 140 | 105 | 4x19 |
| 50 | 3.7 | 4.1 | 6.8 | 130 | 105 | 4x15 | 155 | 120 | 4x19 | 155 | 120 | 8x19 |
| 65 | 4 | 5.5 | 8.9 | 155 | 130 | 4x15 | 175 | 140 | 4x19 | 175 | 140 | 8x19 |
| 80 | 4.4 | 6.8 | 10.2 | 180 | 145 | 4x19 | 185 | 150 | 8x19 | 200 | 160 | 8x23 |
| 100 | 5.9 | 8.6 | 14.5 | 200 | 165 | 8x19 | 210 | 175 | 8x19 | 225 | 185 | 8x23 |
| 125 | 8.9 | 10.5 | 16.8 | 235 | 200 | 8x19 | 250 | 210 | 8x23 | 270 | 225 | 8x25 |
| 150 | 9 | 12.5 | 19.8 | 265 | 230 | 8x19 | 280 | 240 | 8x23 | 305 | 260 | 12x25 |
| 200 | 11 | 21.4 | 31.2 | 320 | 280 | 8x23 | 330 | 290 | 12x23 | 350 | 305 | 12x25 |
| 250 | 20 | 29.3 | 46.2 | 385 | 345 | 12x23 | 400 | 355 | 12x25 | 430 | 380 | 12x27 |
| 300 | 32 | 44 | 58.8 | 430 | 390 | 12x23 | 445 | 400 | 16x25 | 480 | 430 | 16x27 |
| 350 | 43 | 62.2 | 67 | 480 | 435 | 12x25 | 490 | 445 | 16x25 | 540 | 480 | 16x33 |
| 400 | 64 | 112 | 96 | 540 | 495 | 16x25 | 560 | 510 | 16x27 | 605 | 540 | 16x33 |
| 450 | 99 | 153 | 147 | 605 | 555 | 16x25 | 620 | 565 | 20x27 | 675 | 605 | 20x33 |
| 500 | 115 | 199 | 161 | 655 | 605 | 20x25 | 675 | 620 | 20x27 | 730 | 660 | 20x33 |
| 600 | 215 | 283 | 292 | 770 | 715 | 20x27 | 795 | 730 | 24x33 | 845 | 770 | 24x39 |
| 700 | 377 | 490 | 539 | 875 | 820 | 24x27 | 905 | 840 | 24x33 | 960 | 875 | 24x42 |
| 750 | 460 | 598 | 658 | 945 | 880 | 24x33 | 970 | 900 | 24x33 | 1020 | 935 | 24x42 |
| 800 | 542 | 677 | 745 | 995 | 930 | 24x33 | 1020 | 950 | 28x33 | 1085 | 990 | 24x48 |
| 900 | 699 | 874 | 944 | 1095 | 1030 | 24x33 | 1120 | 1050 | 28x33 | 1185 | 1090 | 28x48 |
| 1000 | 977 | 1221 | 1307 | 1195 | 1130 | 28x33 | 1235 | 1160 | 28x39 | 1320 | 1210 | 28x56 |

# JIS B2220 - Flanges 5K & 10K

Available in Carbon Steel and 316L Stainless Steel

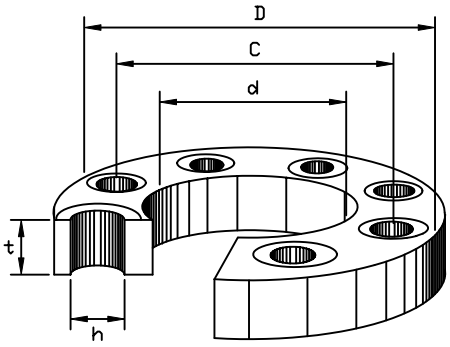


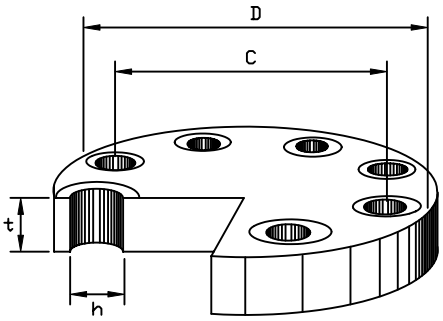




| | JIS 5K | | | | | | | | JIS 10K | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Weight (kg) | | | | | | | | Weight (kg) | |
| Size | d | D | C | No. | h | t | SO | BL | d | D | C | No. | h | t | SO | BL |
| 010 | 17.8 | 75 | 55 | 4 | 12 | 9 | 0.27 | 0.30 | 17.8 | 90 | 65 | 4 | 15 | 12 | 0.52 | 0.54 |
| 015 | 22.2 | 80 | 60 | 4 | 12 | 9 | 0.30 | 0.40 | 22.2 | 95 | 70 | 4 | 15 | 12 | 0.57 | 0.61 |
| 020 | 27.7 | 85 | 65 | 4 | 12 | 10 | 0.37 | 0.50 | 27.7 | 100 | 75 | 4 | 15 | 14 | 0.73 | 0.79 |
| 025 | 34.5 | 95 | 75 | 4 | 12 | 10 | 0.45 | 0.60 | 34.5 | 125 | 90 | 4 | 19 | 14 | 1.13 | 1.24 |
| 032 | 43.2 | 115 | 90 | 4 | 15 | 12 | 0.78 | 0.90 | 43.2 | 135 | 100 | 4 | 19 | 16 | 1.48 | 1.66 |
| 040 | 49.1 | 120 | 95 | 4 | 15 | 12 | 0.83 | 1.00 | 49.1 | 140 | 105 | 4 | 19 | 16 | 1.56 | 1.81 |
| 050 | 61.1 | 130 | 105 | 4 | 15 | 14 | 1.07 | 1.40 | 61.1 | 155 | 120 | 4 | 19 | 16 | 1.88 | 2.23 |
| 065 | 77.1 | 155 | 130 | 4 | 15 | 14 | 1.49 | 2.00 | 77.1 | 175 | 140 | 4 | 19 | 18 | 2.60 | 3.24 |
| 080 | 90.0 | 180 | 145 | 4 | 19 | 14 | 1.99 | 2.70 | 90.0 | 185 | 150 | 8 | 19 | 18 | 2.61 | 3.50 |
| 090 | 102.6 | 190 | 155 | 4 | 19 | 14 | 2.09 | 3.00 | 102.6 | 195 | 160 | 8 | 19 | 18 | 2.76 | 4.00 |
| 100 | 115.4 | 200 | 165 | 8 | 19 | 16 | 2.39 | 3.70 | 115.4 | 210 | 175 | 8 | 19 | 18 | 3.14 | 4.60 |
| 125 | 141.2 | 235 | 200 | 8 | 19 | 16 | 3.23 | 5.20 | 141.2 | 250 | 210 | 8 | 23 | 20 | 4.77 | 7.30 |
| 150 | 166.6 | 265 | 230 | 8 | 19 | 18 | 4.41 | 7.50 | 166.6 | 280 | 240 | 8 | 23 | 22 | 6.34 | 10.10 |
| 175 | 192.1 | 300 | 260 | 8 | 23 | 18 | 5.51 | 9.50 | 192.1 | 302 | 265 | 12 | 23 | 22 | 6.82 | 11.80 |
| 200 | 218.0 | 320 | 280 | 8 | 23 | 20 | 6.33 | 12.20 | 218.0 | 330 | 290 | 12 | 23 | 22 | 7.53 | 14.00 |
| 250 | 269.5 | 385 | 345 | 12 | 23 | 22 | 9.45 | 19.30 | 269.5 | 400 | 355 | 12 | 25 | 24 | 11.80 | 22.70 |
| 300 | 321.0 | 430 | 390 | 12 | 23 | 22 | 10.30 | 24.30 | 321.0 | 445 | 400 | 16 | 25 | 24 | 13.50 | 29.50 |
| 350 | 358.1 | 480 | 435 | 12 | 25 | 24 | 14.00 | 33.20 | 358.1 | 490 | 445 | 16 | 25 | 26 | 16.40 | 37.10 |
| 400 | 409.0 | 540 | 495 | 16 | 25 | 24 | 16.90 | 41.90 | 409.0 | 560 | 510 | 16 | 27 | 28 | 23.00 | 52.50 |
| 450 | 460.0 | 605 | 555 | 16 | 25 | 24 | 21.60 | 53.00 | 460.0 | 620 | 565 | 20 | 27 | 30 | 29.50 | 68.80 |
| 500 | 511.0 | 655 | 605 | 20 | 25 | 24 | 23.10 | 61.90 | 511.0 | 675 | 620 | 20 | 27 | 30 | 33.50 | 82.10 |
| 550 | 562.0 | 720 | 665 | 20 | 27 | 26 | 30.30 | 81.20 | 562.0 | 745 | 680 | 20 | 33 | 32 | 43.10 | 105.80 |
| 600 | 613.0 | 770 | 715 | 20 | 27 | 26 | 32.70 | 93.20 | 613.0 | 795 | 730 | 24 | 33 | 32 | 45.70 | 120.20 |
| 650 | 664.0 | 825 | 770 | 24 | 27 | 26 | 35.90 | 106.90 | 664.0 | 845 | 780 | 24 | 33 | 34 | 52.10 | 145.00 |
| 700 | 715.0 | 875 | 820 | 24 | 27 | 26 | 38.20 | 120.60 | 715.0 | 905 | 840 | 24 | 33 | 34 | 59.50 | 167.20 |
| 750 | 766.0 | 945 | 880 | 24 | 33 | 28 | 48.70 | 150.50 | 766.0 | 970 | 900 | 24 | 33 | 36 | 73.20 | 204.20 |
| 800 | 817.0 | 995 | 930 | 24 | 33 | 28 | 51.60 | 167.40 | 817.0 | 1020 | 950 | 28 | 33 | 36 | 76.40 | 225.20 |
| 900 | 919.0 | 1095 | 1030 | 24 | 33 | 30 | 61.10 | 218.10 | 919.0 | 1120 | 1050 | 28 | 33 | 38 | 89.40 | 288.40 |
| 1000 | 1021.0 | 1195 | 1130 | 28 | 33 | 32 | 70.50 | 277.30 | 1021.0 | 1235 | 1160 | 28 | 39 | 40 | 109.20 | 367.70 |

# JIS B2220 - Flanges 16K & 20K

Available in Carbon Steel and 316L Stainless Steel




| | JIS 16K | | | | | | | | JIS 20K | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Weight (kg) | | | | | | | | Weight (kg) | |
| Size | d | D | C | no | h | t | SO | BL | d | D | C | no | h | t | SO | BL |
| 010 | 17.8 | 90 | 65 | 4 | 15 | 12 | 0.51 | 0.53 | 17.8 | 90 | 65 | 4 | 15 | 14 | 0.60 | 0.60 |
| 015 | 22.2 | 95 | 70 | 4 | 15 | 12 | 0.56 | 0.60 | 22.2 | 95 | 70 | 4 | 15 | 14 | 0.70 | 0.70 |
| 020 | 27.7 | 100 | 75 | 4 | 15 | 14 | 0.72 | 0.79 | 27.7 | 100 | 75 | 4 | 15 | 16 | 0.80 | 0.90 |
| 025 | 34.5 | 125 | 90 | 4 | 19 | 14 | 1.13 | 1.23 | 34.5 | 125 | 90 | 4 | 19 | 16 | 1.30 | 1.50 |
| 032 | 43.2 | 135 | 100 | 4 | 19 | 16 | 1.46 | 1.66 | 43.2 | 135 | 100 | 4 | 19 | 18 | 1.60 | 1.80 |
| 040 | 49.1 | 140 | 105 | 4 | 19 | 16 | 1.56 | 1.81 | 49.1 | 140 | 105 | 4 | 19 | 18 | 1.70 | 2.00 |
| 050 | 61.1 | 155 | 120 | 8 | 19 | 16 | 1.70 | 2.10 | 61.1 | 155 | 120 | 8 | 19 | 18 | 1.90 | 2.40 |
| 065 | 77.1 | 175 | 140 | 8 | 19 | 18 | 2.40 | 3.10 | 77.1 | 175 | 140 | 8 | 19 | 20 | 2.60 | 3.40 |
| 080 | 90.0 | 200 | 160 | 8 | 23 | 20 | 3.50 | 4.50 | 90.0 | 200 | 160 | 8 | 23 | 22 | 3.80 | 4.90 |
| 090 | 102.6 | 210 | 170 | 8 | 23 | 20 | 3.70 | 5.00 | 102.6 | 210 | 170 | 8 | 23 | 24 | 4.40 | 6.00 |
| 100 | 115.4 | 225 | 185 | 8 | 23 | 22 | 4.50 | 6.30 | 115.4 | 225 | 185 | 8 | 23 | 24 | 4.90 | 6.90 |
| 125 | 141.2 | 270 | 225 | 8 | 25 | 22 | 6.50 | 9.20 | 141.2 | 270 | 225 | 8 | 25 | 26 | 7.80 | 11.00 |
| 150 | 166.6 | 305 | 260 | 12 | 25 | 24 | 8.70 | 12.80 | 166.6 | 305 | 260 | 12 | 25 | 28 | 10.10 | 14.90 |
| 175 | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| 200 | 218.0 | 350 | 305 | 12 | 25 | 26 | 10.90 | 18.60 | 218.0 | 350 | 305 | 12 | 25 | 30 | 12.60 | 21.40 |
| 250 | 269.5 | 430 | 380 | 12 | 27 | 28 | 18.00 | 30.60 | 269.5 | 430 | 380 | 12 | 27 | 34 | 21.90 | 37.20 |
| 300 | 321.0 | 480 | 430 | 16 | 27 | 30 | 21.50 | 40.70 | 321.0 | 480 | 430 | 16 | 27 | 36 | 25.80 | 48.80 |
| 350 | 358.1 | 540 | 480 | 16 | 33 | 34 | 30.80 | 57.80 | 358.1 | 540 | 480 | 16 | 33 | 40 | 36.20 | 68.00 |
| 400 | 409.0 | 605 | 540 | 16 | 33 | 38 | 42.80 | 82.20 | 409.0 | 605 | 540 | 16 | 33 | 46 | 51.70 | 99.40 |
| 450 | 460.0 | 675 | 605 | 20 | 33 | 40 | 55.10 | 107.60 | 460.0 | 675 | 605 | 20 | 33 | 48 | 66.10 | 129.10 |
| 500 | 511.0 | 730 | 660 | 20 | 33 | 42 | 65.10 | 133.10 | 511.0 | 730 | 660 | 20 | 33 | 50 | 77.40 | 158.40 |
| 550 | 562.0 | 795 | 720 | 20 | 39 | 44 | 77.90 | 164.10 | 562.0 | 795 | 720 | 20 | 39 | 52 | 92.20 | 194.00 |
| 600 | 613.0 | 845 | 770 | 24 | 39 | 46 | 86.00 | 193.20 | 613.0 | 845 | 770 | 24 | 39 | 54 | 101.10 | 226.90 |
| 650 | 664.0 | 895 | 820 | 24 | 39 | 48 | 96.30 | 227.50 | 664.0 | 945 | 850 | 24 | 48 | 60 | 147.60 | 311.60 |
| 700 | 715.0 | 960 | 875 | 24 | 42 | 50 | 114.10 | 272.60 | 715.0 | 995 | 900 | 24 | 48 | 64 | 168.00 | 370.90 |
| 750 | 766.0 | 1020 | 935 | 24 | 42 | 52 | 132.70 | 321.90 | 766.0 | 1080 | 970 | 24 | 56 | 68 | 212.70 | 460.10 |
| 800 | 817.0 | 1085 | 990 | 24 | 48 | 54 | 152.10 | 375.60 | 817.0 | 1140 | 1030 | 24 | 56 | 72 | 248.50 | 546.50 |
| 900 | 919.0 | 1185 | 1090 | 28 | 48 | 58 | 178.10 | 481.80 | 919.0 | 1250 | 1140 | 28 | 56 | 76 | 296.90 | 694.90 |
| 1000 | 1021.0 | 1320 | 1210 | 28 | 56 | 62 | 235.30 | 636.00 | - | - | - | - | - | - | - | - |

WORLD WIDE METRIC

# Conversion Table - Linear

From Inches to Millimeters

1 Inch = 25.4mm

| Inch | 0 | 1/16 | 1/8 | 3/16 | 1/4 | 5/16 | 3/8 | 7/16 | 1/2 | 9/16 | 5/8 | 11/16 | 3/4 | 13/16 | 7/8 | 15/16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | mm | 1.6 | 3.2 | 4.8 | 6.4 | 7.9 | 9.5 | 11.1 | 12.7 | 14.3 | 15.9 | 17.5 | 19.1 | 20.6 | 22.2 | 23.8 |
| 1 | 25.4 | 27.0 | 28.6 | 30.2 | 31.8 | 33.3 | 34.9 | 36.5 | 38.1 | 39.7 | 41.3 | 42.9 | 44.5 | 46.0 | 47.6 | 49.2 |
| 2 | 50.8 | 52.4 | 54.0 | 55.6 | 57.2 | 58.7 | 60.3 | 61.9 | 63.5 | 65.1 | 66.7 | 68.3 | 69.9 | 71.4 | 73.0 | 74.6 |
| 3 | 76.2 | 77.8 | 79.4 | 81.0 | 82.6 | 84.1 | 85.7 | 87.3 | 88.9 | 90.5 | 92.1 | 93.7 | 95.3 | 96.8 | 98.4 | 100.0 |
| 4 | 101.6 | 103.2 | 104.8 | 106.4 | 108.0 | 109.5 | 111.1 | 112.7 | 114.3 | 115.9 | 117.5 | 119.1 | 120.7 | 122.2 | 123.8 | 125.4 |
| 5 | 127.0 | 128.6 | 130.2 | 131.8 | 133.4 | 134.9 | 136.5 | 138.1 | 139.7 | 141.3 | 142.9 | 144.5 | 146.1 | 147.6 | 149.2 | 150.8 |
| 6 | 152.4 | 154.0 | 155.6 | 157.2 | 158.8 | 160.3 | 161.9 | 163.5 | 165.1 | 166.7 | 168.3 | 169.9 | 171.5 | 173.0 | 174.6 | 176.2 |
| 7 | 177.8 | 179.4 | 181.0 | 182.6 | 184.2 | 185.7 | 187.3 | 188.9 | 190.5 | 192.1 | 193.7 | 195.3 | 196.9 | 198.4 | 200.0 | 201.6 |
| 8 | 203.2 | 204.8 | 206.4 | 208.0 | 209.6 | 211.1 | 212.7 | 214.3 | 215.9 | 217.5 | 219.1 | 220.7 | 222.3 | 223.8 | 225.4 | 227.0 |
| 9 | 228.6 | 230.2 | 231.8 | 233.4 | 235.0 | 236.5 | 238.1 | 239.7 | 241.3 | 242.9 | 244.5 | 246.1 | 247.7 | 249.2 | 250.8 | 252.4 |
| 10 | 254.0 | 255.6 | 257.2 | 258.8 | 260.4 | 261.9 | 263.5 | 265.1 | 266.7 | 268.3 | 269.9 | 271.5 | 273.1 | 274.6 | 276.2 | 277.8 |
| 11 | 279.4 | 281.0 | 282.6 | 284.2 | 285.8 | 287.3 | 288.9 | 290.5 | 292.1 | 293.7 | 295.3 | 296.9 | 298.5 | 300.0 | 301.6 | 303.2 |
| 12 | 304.8 | 306.4 | 308.0 | 309.6 | 311.2 | 312.7 | 314.3 | 315.9 | 317.5 | 319.1 | 320.7 | 322.3 | 323.9 | 325.4 | 327.0 | 328.6 |
| 13 | 330.2 | 331.8 | 333.4 | 335.0 | 336.6 | 338.1 | 339.7 | 341.3 | 342.9 | 344.5 | 346.1 | 347.7 | 349.3 | 350.8 | 352.4 | 354.0 |
| 14 | 355.6 | 357.2 | 358.8 | 360.4 | 362.0 | 363.5 | 365.1 | 366.7 | 368.3 | 369.9 | 371.5 | 373.1 | 374.7 | 376.2 | 377.8 | 379.4 |
| 15 | 381.0 | 382.6 | 384.2 | 385.8 | 387.4 | 388.9 | 390.5 | 392.1 | 393.7 | 395.3 | 396.9 | 398.5 | 400.1 | 401.6 | 403.2 | 404.8 |
| 16 | 406.4 | 408.0 | 409.6 | 411.2 | 412.8 | 414.3 | 415.9 | 417.5 | 419.1 | 420.7 | 422.3 | 423.9 | 425.5 | 427.0 | 428.6 | 430.2 |
| 17 | 431.8 | 433.4 | 435.0 | 436.6 | 438.2 | 439.7 | 441.3 | 442.9 | 444.5 | 446.1 | 447.7 | 449.3 | 450.9 | 452.4 | 454.0 | 455.6 |
| 18 | 457.2 | 458.8 | 460.4 | 462.0 | 463.6 | 465.1 | 466.7 | 468.3 | 469.9 | 471.5 | 473.1 | 474.7 | 476.3 | 477.8 | 479.4 | 481.0 |
| 19 | 482.6 | 484.2 | 485.8 | 487.4 | 489.0 | 490.5 | 492.1 | 493.7 | 495.3 | 496.9 | 498.5 | 500.1 | 501.7 | 503.2 | 504.8 | 506.4 |
| 20 | 508.0 | 509.6 | 511.2 | 512.8 | 514.4 | 515.9 | 517.5 | 519.1 | 520.7 | 522.3 | 523.9 | 525.5 | 527.1 | 528.6 | 530.2 | 531.8 |
| 21 | 533.4 | 535.0 | 536.6 | 538.2 | 539.8 | 541.3 | 542.9 | 544.5 | 546.1 | 547.7 | 549.3 | 550.9 | 552.5 | 554.0 | 555.6 | 557.2 |
| 22 | 558.8 | 560.4 | 562.0 | 563.6 | 565.2 | 566.7 | 568.3 | 569.9 | 571.5 | 573.1 | 574.7 | 576.3 | 577.9 | 579.4 | 581.0 | 582.6 |
| 23 | 584.2 | 585.8 | 587.4 | 589.0 | 590.6 | 592.1 | 593.7 | 595.3 | 596.9 | 598.5 | 600.1 | 601.7 | 603.3 | 604.8 | 606.4 | 608.0 |
| 24 | 609.6 | 611.2 | 612.8 | 614.4 | 616.0 | 617.5 | 619.1 | 620.7 | 622.3 | 623.9 | 625.5 | 627.1 | 628.7 | 630.2 | 631.8 | 633.4 |
| 25 | 635.0 | 636.6 | 638.2 | 639.8 | 641.4 | 642.9 | 644.5 | 646.1 | 647.7 | 649.3 | 650.9 | 652.5 | 654.1 | 655.6 | 657.2 | 658.8 |

Example:
In order to obtain the millimeter (mm) dimension for 7-1/2”, find 7” on the x (vertical) axis and 1/2” on the y (horizontal) axis of the chart. The intersection of the two numbers is the millimeter (mm) equivalent to 7-1/2” (or 190.50mm).

WORLD WIDE METRIC

# Conversion Table - Pressure

From PSI to Kg/Cm2 and Kg/Cm2 to PSI

| PSI | Kg/ Cm2 | PSI | Kg/ Cm2 | PSI | Kg/ Cm2 | PSI | Kg/ Cm2 | PSI | Kg/ Cm2 | PSI | Kg/ Cm2 | PSI | Kg/ Cm2 | PSI | Kg/ Cm2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.07 | 31 | 2.18 | 61 | 4.29 | 91 | 6.40 | 205 | 14.42 | 410 | 28.83 | 710 | 49.93 | 1010 | 71.03 |
| 2 | 0.14 | 32 | 2.25 | 62 | 4.36 | 92 | 6.47 | 210 | 14.77 | 420 | 29.54 | 720 | 50.63 | 1020 | 71.73 |
| 3 | 0.21 | 33 | 2.32 | 63 | 4.43 | 93 | 6.54 | 215 | 15.12 | 430 | 30.24 | 730 | 51.34 | 1030 | 72.43 |
| 4 | 0.28 | 34 | 2.39 | 64 | 4.50 | 94 | 6.61 | 220 | 15.47 | 440 | 30.94 | 740 | 52.04 | 1040 | 73.14 |
| 5 | 0.35 | 35 | 2.46 | 65 | 4.57 | 95 | 6.68 | 225 | 15.82 | 450 | 31.65 | 750 | 52.74 | 1050 | 73.84 |
| 6 | 0.42 | 36 | 2.53 | 66 | 4.64 | 96 | 6.75 | 230 | 16.17 | 460 | 32.35 | 760 | 53.45 | 1060 | 74.54 |
| 7 | 0.49 | 37 | 2.60 | 67 | 4.71 | 97 | 6.82 | 235 | 16.53 | 470 | 33.05 | 770 | 54.15 | 1070 | 75.25 |
| 8 | 0.56 | 38 | 2.67 | 68 | 4.78 | 98 | 6.89 | 240 | 16.88 | 480 | 33.76 | 780 | 54.85 | 1080 | 75.95 |
| 9 | 0.63 | 39 | 2.74 | 69 | 4.85 | 99 | 6.96 | 245 | 17.23 | 490 | 34.46 | 790 | 55.56 | 1090 | 76.65 |
| 10 | 0.70 | 40 | 2.81 | 70 | 4.92 | 100 | 7.03 | 250 | 17.58 | 500 | 35.16 | 800 | 56.26 | 1100 | 77.36 |
| 11 | 0.77 | 41 | 2.88 | 71 | 4.99 | 105 | 7.38 | 255 | 17.93 | 510 | 35.86 | 810 | 56.96 | 1120 | 78.76 |
| 12 | 0.84 | 42 | 2.95 | 72 | 5.06 | 110 | 7.74 | 260 | 18.28 | 520 | 36.57 | 820 | 57.67 | 1140 | 80.17 |
| 13 | 0.91 | 43 | 3.02 | 73 | 5.13 | 115 | 8.09 | 265 | 18.64 | 530 | 37.27 | 830 | 58.37 | 1160 | 81.58 |
| 14 | 0.98 | 44 | 3.09 | 74 | 5.20 | 120 | 8.44 | 270 | 18.99 | 540 | 37.97 | 840 | 59.07 | 1180 | 82.98 |
| 15 | 1.05 | 45 | 3.16 | 75 | 5.27 | 125 | 8.79 | 275 | 19.34 | 550 | 38.68 | 850 | 59.77 | 1200 | 84.39 |
| 16 | 1.13 | 46 | 3.23 | 76 | 5.34 | 130 | 9.14 | 280 | 19.69 | 560 | 39.38 | 860 | 60.48 | 1220 | 85.79 |
| 17 | 1.20 | 47 | 3.31 | 77 | 5.41 | 135 | 9.49 | 285 | 20.04 | 570 | 40.08 | 870 | 61.18 | 1240 | 87.20 |
| 18 | 1.27 | 48 | 3.38 | 78 | 5.49 | 140 | 9.85 | 290 | 20.39 | 580 | 40.79 | 880 | 61.88 | 1260 | 88.61 |
| 19 | 1.34 | 49 | 3.45 | 79 | 5.56 | 145 | 10.20 | 295 | 20.75 | 590 | 41.49 | 890 | 62.59 | 1280 | 90.01 |
| 20 | 1.41 | 50 | 3.52 | 80 | 5.63 | 150 | 10.55 | 300 | 21.10 | 600 | 42.19 | 900 | 63.29 | 1300 | 91.42 |
| 21 | 1.48 | 51 | 3.59 | 81 | 5.70 | 155 | 10.90 | 310 | 21.80 | 610 | 42.90 | 910 | 63.99 | 1320 | 92.83 |
| 22 | 1.55 | 52 | 3.66 | 82 | 5.77 | 160 | 11.25 | 320 | 22.50 | 620 | 43.60 | 920 | 64.70 | 1340 | 94.23 |
| 23 | 1.62 | 53 | 3.73 | 83 | 5.84 | 165 | 11.60 | 330 | 23.21 | 630 | 44.30 | 930 | 65.40 | 1360 | 95.64 |
| 24 | 1.69 | 54 | 3.80 | 84 | 5.91 | 170 | 11.95 | 340 | 23.91 | 640 | 45.01 | 940 | 66.10 | 1380 | 97.05 |
| 25 | 1.76 | 55 | 3.87 | 85 | 5.98 | 175 | 12.31 | 350 | 24.61 | 650 | 45.71 | 950 | 66.81 | 1400 | 98.45 |
| 26 | 1.83 | 56 | 3.94 | 86 | 6.05 | 180 | 12.66 | 360 | 25.32 | 660 | 46.41 | 960 | 67.51 | 1420 | 99.86 |
| 27 | 1.90 | 57 | 4.01 | 87 | 6.12 | 185 | 13.01 | 370 | 26.02 | 670 | 47.12 | 970 | 68.21 | 1440 | 101.27 |
| 28 | 1.97 | 58 | 4.08 | 88 | 6.19 | 190 | 13.36 | 380 | 26.72 | 680 | 47.82 | 980 | 68.92 | 1460 | 102.67 |
| 29 | 2.04 | 59 | 4.15 | 89 | 6.26 | 195 | 13.71 | 390 | 27.43 | 690 | 48.52 | 990 | 69.62 | 1480 | 104.08 |
| 30 | 2.11 | 60 | 4.22 | 90 | 6.33 | 200 | 14.06 | 400 | 28.13 | 700 | 49.23 | 1000 | 70.32 | 1500 | 105.49 |

To convert to Kg/Cm2 from PSI, divide PSI by 14.22

To convert to PSI from Kg/Cm2, multiply Kg/Cm2 by 14.22

WORLD WIDE METRIC

# Conversion Table - Temperature

From Fahreneheit to Celsuis and Celsuis to Fahrenheit

| F | C | F | C | F | C | F | C | F | C | F | C |
|---|---|---|---|---|---|---|---|---|---|---|---|
| -160 | -106.7 | -40 | -40.0 | 75 | 23.9 | 195 | 90.6 | 850 | 454.4 | 2050 | 1121.1 |
| -155 | -103.9 | -35 | -37.2 | 80 | 26.7 | 200 | 93.3 | 900 | 482.2 | 2100 | 1148.9 |
| -150 | -101.1 | -30 | -34.4 | 85 | 29.4 | 205 | 96.1 | 950 | 510.0 | 2150 | 1176.7 |
| -145 | -98.3 | -25 | -31.7 | 90 | 32.2 | 210 | 98.9 | 1000 | 537.8 | 2200 | 1204.4 |
| -140 | -95.6 | -20 | -28.9 | 95 | 35.0 | 212 | 100.0 | 1050 | 565.6 | 2250 | 1232.2 |
| -135 | -92.8 | -15 | -26.1 | 100 | 37.8 | 215 | 101.7 | 1100 | 593.3 | 2300 | 1260.0 |
| -130 | -90.0 | -10 | -23.3 | 105 | 40.6 | 220 | 104.4 | 1150 | 621.1 | 2350 | 1287.8 |
| -125 | -87.2 | -5 | -20.6 | 110 | 43.3 | 225 | 107.2 | 1200 | 648.9 | 2400 | 1315.6 |
| -120 | -84.4 | 0 | -17.8 | 115 | 46.1 | 230 | 110.0 | 1250 | 676.7 | 2450 | 1343.3 |
| -115 | -81.7 | 5 | -15.0 | 120 | 48.9 | 235 | 112.8 | 1300 | 704.4 | 2500 | 1371.1 |
| -110 | -78.9 | 10 | -12.2 | 125 | 51.7 | 240 | 115.6 | 1350 | 732.2 | 2550 | 1398.9 |
| -105 | -76.1 | 15 | -9.4 | 130 | 54.4 | 245 | 118.3 | 1400 | 760.0 | 2600 | 1426.7 |
| -100 | -73.3 | 20 | -6.7 | 135 | 57.2 | 250 | 121.1 | 1450 | 787.8 | 2650 | 1454.4 |
| -95 | -70.6 | 25 | -3.9 | 140 | 60.0 | 300 | 148.9 | 1500 | 815.6 | 2700 | 1482.2 |
| -90 | -67.8 | 30 | -1.1 | 145 | 62.8 | 350 | 176.7 | 1550 | 843.3 | 2750 | 1510.0 |
| -85 | -65.0 | 32 | 0.0 | 150 | 65.6 | 400 | 204.4 | 1600 | 871.1 | 2800 | 1537.8 |
| -80 | -62.2 | 35 | 1.7 | 155 | 68.3 | 450 | 232.2 | 1650 | 898.9 | 2850 | 1565.6 |
| -75 | -59.4 | 40 | 4.4 | 160 | 71.1 | 500 | 260.0 | 1700 | 926.7 | 2900 | 1593.3 |
| -70 | -56.7 | 45 | 7.2 | 165 | 73.9 | 550 | 287.8 | 1750 | 954.4 | 2950 | 1621.1 |
| -65 | -53.9 | 50 | 10.0 | 170 | 76.7 | 600 | 315.6 | 1800 | 982.2 | 3000 | 1648.9 |
| -60 | -51.1 | 55 | 12.8 | 175 | 79.4 | 650 | 343.3 | 1850 | 1010.0 | 3050 | 1676.7 |
| -55 | -48.3 | 60 | 15.6 | 180 | 82.2 | 700 | 371.1 | 1900 | 1037.8 | 3100 | 1704.4 |
| -50 | -45.6 | 65 | 18.3 | 185 | 85.0 | 750 | 398.9 | 1950 | 1065.6 | 3150 | 1732.2 |
| -45 | -42.8 | 70 | 21.1 | 190 | 87.8 | 800 | 426.7 | 2000 | 1093.3 | 3200 | 1760.0 |

To convert to Fahrenheit from Celsuis: $F = 32 + 9/5\ C$

To convert to Celsuis from Fahrenheit: $C = 5/9\ (F-32)$